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Technical Report No. 86

Industry Challenges and Current Technologies for Pharmaceutical Package Integrity Testing

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1.0 Introduction

For more than a decade, PDA *Technical Report No. 27: Pharmaceutical Package Integrity* served as an influential reference document, representing the best practices in sterile product package integrity testing. At the time it was published, the information it presented was instrumental in shaping the revision of *USP General Chapter <1207> Package Integrity and Test Method Selection* ***(1, 2)***. As a result, the content of USP <1207> was expanded in 2016 and currently serves as the primary reference chapter focusing on the basics of package integrity and best practices in sterile product package integrity evaluation and integrity testing. The intent of USP <1207> is to guide package integrity development away from testing each sample at the smallest possible defect level, because quality cannot be tested into a product. Rather, using a risk-based approach and leveraging testing to better understand, analyze, and eliminate the risks during development phases is encouraged. Use of the appropriate testing during each manufacturing phase can help in evaluating and mitigating residual risks (i.e., in production/stability to proactively address risks/failure modes). Where feasible, regulations and quality risk management principles identified in ICH Quality Guideline Q9: Quality Risk Management should be used ***(3)***.

Subsequently, the PDA Package Integrity Technical Report Team reexamined the role of TR-27 and determined that a new technical report should be written, building on the package integrity basics updated by USP <1207> and focusing more on advanced issues related to pharmaceutical package integrity. They researched the existing sources of information related to the challenges encountered with complex packaging systems, such as large-volume bulk packages and prefilled syringe systems. Currently, no single source is available that thoroughly addresses these issues. Likewise, the Technical Report Team recognized the increasing number of innovative methods using existing technologies for integrity testing that deserve consideration for their potential utility in the future of pharmaceutical package integrity.

This technical report is a consensus-based resource surrounding the challenges encountered in using complex package systems and introduces important elements to consider in decision-making. It also offers an examination of the technologies available for package integrity testing not yet established by peer-reviewed research.

Although pharmaceutical container closure systems serve a variety of functions, the assurance of package integrity is primarily concerned with product containment. However, these systems also serve as an effective barrier against the ingress of microorganisms and potentially reactive gases, such as oxygen or water vapor, and sometimes function to maintain a vacuum.

Package integrity is the measure of the ability of the package to prevent product loss or maintain product sterility and the ability to maintain the internal environment (e.g., loss of vacuum seal). Package integrity testing is an ongoing process that starts in package development and continues throughout the life of the product. When designing a container closure system and selecting and implementing an appropriate method, the goal is to reduce the risk to product sterility and quality.

Best practices in pharmaceutical packaging suggest that all activities be evaluated for risk to the product quality as a surrogate to patient safety. Failure of container closure integrity is a critical risk, as the lack of package integrity can lead to contamination of the product which can impact both product quality and potency. Contamination of a sterile product intended for parenteral administration could result in dire consequences for the patient.

Demonstration of container closure integrity and ensuring the integrity of the container closure system is required throughout the lifecycle of a sterile pharmaceutical product. A well-characterized and controlled process should consist of a qualified container closure design with effective, monitored process controls in place; this ensures the process parameters required to produce integral container closure systems are maintained. These controls should establish appropriate specifications for incoming components, qualification of manufacturing equipment, including a visual inspection system for detection of irregularities in the process, and appropriate sampling strategies for container closure integrity testing.

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Package integrity should ensure product quality is maintained through both the shelf life and life of a product. Over the past two decades, analytical technology has progressed, improving the ability to evaluate closure integrity. The information provided in this technical report is intended to assist users in advancing their current integrity assessment and testing strategies during the phases of product lifecycle. The report addresses container closure integrity in three main areas: challenges with complex systems and package integrity methods, innovative methods for integrity testing using existing technologies, and additional considerations for pharmaceutical package integrity. Additionally, how quality by design (QbD) relates to package integrity and establishing a limit of detection are discussed in the final section.

1.1 Purpose and Scope

This technical report focuses on the challenges facing the pharmaceutical industry that use complex packaging systems for sterile drugs and biologics (e.g., syringes, syringe assemblies, bulk containers). It also presents information on some innovative methods for package integrity testing using existing technologies, including the potential impact of cryogenic conditions. The intent is to update information and incorporate experiential learning which is not addressed in PDA TR-27 and USP <1207> ***(1,2)***.

It also serves as a technical resource, focusing particularly on sterile products and encouraging a risk-based approach and leveraging testing to better understand, analyze, and eliminate the risks during developmental phases. Use of the appropriate testing during each manufacturing phase can help in evaluating and mitigating residual risks.

The information in this technical report is the result of a consensus achieved by the PDA Container Closure Integrity Technical Report Team and are not necessarily the views of regulatory agencies.

2.0 Glossary

Definitions have been provided to help clarify the concepts discussed in this technical report. While some of these definitions vary among companies, the definitions described below are consistent for use within this document.

Container Closure System (CCS)

The sum of packaging components (primary and secondary) and materials that together contain and protect a product.

Container Closure Integrity Test (CCIT)

A package leak test (either physicochemical or microbiological) that detects the presence of a package breach or gap. Some tests may also be able to identify the magnitude and/or location of the leak (the term container closure integrity test is synonymous with package leak test or package integrity test for the purposes of this TR).

Destructive

A method in which a sample cannot be subsequently utilized in any other analytical method or processed to a final product.

Deterministic Leak Test Method

A method in which the leakage event being detected or measured is based on phenomena that follow a predictable chain of events. In addition, the measure of leak detection is based on physicochemical technologies that are readily controlled and monitored, yielding objective quantitative data.

Labeling

The process by which a label is affixed to a packaging component.

Maximum Allowable Leak Limit (MALL)

The greatest gap or leak rate that does not put product quality at risk ***(2)***.

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Nondestructive
A method in which a passing sample can subsequently be utilized in additional analytical methods or processed to a final product.

Package integrity
Is the measure of the ability of the package to prevent product loss or maintain product sterility and the ability to maintain the internal environment *(2)*.

Probabilistic Leak Test Method
A method that is the converse of a deterministic leak test method, being stochastic in nature. Probabilistic tests rely on a series of sequential and/or simultaneous events, each associated with random outcomes described by probability distributions.

Sterile/Microbial Barrier
The purported location on a container closure system beyond which no microorganism can pass under conditions qualified for the barrier.

2.1 Abbreviations

CCI Container Closure Integrity

CCIT Container Closure Integrity Test

CCS Container Closure Systems

CPP Critical Process Parameter

CQA Critical Quality Attributes

HVLD High-Voltage Leak Detection

FISTA International Safe Transit Association

MALL Maximum Allowable Leak Limit

LOD Limit of Detection

OES Optical Emission Spectroscopy

QbD Quality by Design

3.0 Challenges with Methodologies

The selection of container closure integrity methodology is based on sensitivity, reduced variability, and clarification of the intended result of leak detection. The selected method will present challenges to the investigator, however, inherent characteristics of the chosen method and other unknown factors can increase variability, for example:

- Formulations that impact flow through a leak or that create blockages within the leak.
- Methods that impact formulations, e.g., a method that applies an energy level or introduces a new chemical compound that impacts the original formulation in any way.
- Unintended impacts of method on a container closure system (CCS) potentially introducing a defect to or masking a defect in an otherwise unadulterated sample, e.g., pressure changing the shape of the container and potentially the nature of the defect.

Section 3 addresses the challenges and potential impact of emerging issues on methodology design and selection of container closure integrity testing (CCIT).

3.1 Positive Controls

Container closure integrity (CCI) methods use positive control units, also called "leak artifacts." These controls are useful for different purposes and during different stages of test methods and product development, such as:

- Design and development of the container
- Development of a test method

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- Determination of the level of sensitivity to various types of leaks
- Determination of detection limits
- Validation of a test method
- Validation or system suitability of the level of sensitivity to various types of leaks
- Testing of containers for their stability properties
- Development of container manufacturing process (quality control) parameters

Naturally occurring leaks are valuable positive controls during qualification of a CCIT method. Leaks that occur naturally also have a limited shelf life; they may be measurable today but not tomorrow. These samples might best be used for real-condition comparison of the method to better characterize the capability of the method to detect actual leak paths when engineered leak paths are used to characterize and develop a method. Acquiring a sufficient set of leak artifacts for development purposes is a challenge due to the difficulty of using the normal container production processes to generate consistent, naturally occurring leaks. Therefore, positive control units with simulated defects can be produced to represent potential container and closure defects to use in method development and verification.

Naturally occurring leaks can also be used during process and product development. In most cases, the geometry and leak size of naturally occurring leaks are unknown, whereas positive controls with simulated defects (such as hole-type defects with channels) usually have leak paths with known geometry that can be generated using a controlled process. Use of both types of positive controls can provide useful information to develop methods for integrity testing.

Some CCIT system instruments can use calibrated leaks, another type of positive control. Calibrated leaks are systematically built into the CCIT system hardware to simulate a leak. The leaks are calibrated against traceable standards and installed inside robust housing (not the product container). If possible, the calibrated leak size should be equal to or less than the maximum allowed leak (MALL) for the specific container design ***(2)***. For a rapid 100% CCIT method, the MALL may not be achieved. The calibrated leaks are used in routine system performance verification and, for some CCIT methods, as tools for routine and frequent calibration. The way in which the leaks were created and calibrated should be completely described.

Note: References that provide a broad overview of historical approaches to defect development and application can be found in the additional reading section.

3.1.1 Considerations for Selecting Positive Controls

Positive controls should be selected based on product package profile, method type, and intended outcome of the method used during the intended lifecycle phase. Positive controls used for method development and validation should include a range of defect sizes and can include a set of different types of defects. At a minimum it should include a set of surrounding leak sizes (above and below) that include the MALL rate for the specific product package and the intended limit of detection for the use of the selected method. There is a limit to these types of defects; therefore, the method cannot be fully characterized to understand its capability or the true MALL of a CCS.

3.1.2 Considerations When Using Positive Controls

Naturally occurring positive controls generated from manufacturing process failures present a challenge when attempting to control the size of leaks. In many cases, these types of defects generate large leaks, or leaks that are fragile and can be easily plugged by product. If found, these types of defects may be valuable to understand method performance against naturally occurring defects but naturally occurring defects should not be used for method validation due to the wide variability they exhibit.

Positive controls with simulated defects can also be easily plugged during handling and storage; therefore, proper storage and handling precautions should be used.

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During the development and validation or qualification of a CCIT method, the test system must be challenged with positive controls. Positive controls should simulate large or gross leaks and those that simulate small leaks. Table 3.1.4-1 offers a summary of common positive controls or engineered defects. Engineered defects for positive controls can have the added advantage of being a micron-sized defect, but other factors should also be considered:

- Percentage of the micron-sized defects that can be consistently and reliably detected using a chosen CCIT method *(NOTE: All positive controls with leaks at or above the claimed limit of detection (LoD) of the test method must be detected with high confidence (e.g., 95%).*
- Size and geometry of leaks that occur in real life and the ability of the test method to accurately detect those defects.
- Leak size needed for microbial contamination to occur and pose a risk to product sterility or quality over the entire lifecycle of the product.

3.1.3 Types of Simulated Positive Controls

USP <1207.1> describes defect types as "positive controls representing realistic package flaws." Along with the examples given, it acknowledges that a chosen CCIT method may not be able to accurately detect some type defects in a given package design, so an alternative method should be identified. When a method cannot be identified, any information and design that can reduce the possible occurrence of such defects in the CCS should be sought ***(4)***.

Common materials used to create simulated defects include:

- Glass micropipette
- Wire
- Laser-drilled

Glass micropipettes are available in diameter sizes from 0.1 µm to 10 µm. The main advantage of a glass micropipette over laser-drilled and wire-simulated defects is the ability to generate relatively consistent defects at small sizes ***(1, 2)***.

Use of a glass micropipette is considered a short microchannel of small diameter (5 µm and under); diameters under 2 µm are not perfectly round.

The micropipette is inserted through a protective tool (such as a hollow metallic syringe needle) into the septum or stopper and bonded to the container with epoxy, or through a hole drilled into the side glass or rigid wall and bonded with epoxy. A microscopic examination of the micropipette tip can be used to verify that the micropipette tip was not broken during sample preparation; then, a test cycle can be performed with an equivalent simulated flow rate for comparison to verify the actual performance of the micropipette ***(1)***.

Glass micropipettes are fragile, and the tip can be easily broken during sample preparation. Broken micropipette tips will typically yield an unusually higher flow rate; such a sample should not be used as a positive control.

Wires can simulate foreign objects, such a human hair (diameter range of 80 µm to 100 µm) at the sealing area. Various wire material and sizes are available in diameters of 10 µm and greater. For some sterile glass containers, a wire can be placed between the plunger and the glass barrel (syringes, cartridges) or between the aluminum-crimped stopper and the glass vial. The elastomeric plunger or stopper, which has been press-fitted to the container, will not fully deform around the wire; therefore, small leak paths can be generated. This type of leak is also considered a microchannel. The defect size is dependent on, for example, the wire size, elastomeric material formulation and hardness, press-fit forces, or assembly forces (crimping for vials). The time between placing the wire and testing may also have some impact due to potential relaxation of elastomeric material. Changes in elastomeric hardness by temperature

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may also have an impact. When these parameters are controlled, repeatable leak paths can be generated. Leak paths with calculated or estimated sizes of 2 µm to 10 µm and less have been created for glass syringes and glass vials using nickel-titanium wires with diameters of 25 µm and greater ***(1, 4)***.

Stainless steel wires (typically 50 µm and larger diameter) are used as simulated defects for heat-sealed or ultrasonically sealed trays and pouches. For lidded trays and pouches, where the container and lid material do not conform well to the wires generating a very large defect, the wire is pulled out shortly after sealing (and before full seal-curing) to generate a smaller leak path. In this case, the strength and width of the seal may not allow for wires of a softer wire material (such as copper) or a smaller diameter to be left intact after pulling.

A laser at various wave lengths, when applied to the container surface, generates localized heat, melting the material and creating an opening (or hole) in the package wall. Laser-drilling on the container walls will generate defects from equivalent sizes of 2 µm and greater. Containers are typically filled and sealed after the laser-drilling process ***(1)***.

Laser-drilling of glass products (e.g., vials, cartridges, and syringes) is done by drilling a tapered hole in the glass, where the hole does not protrude through the glass wall. The heat generated by this process results in cracking the glass near that hole, creating the leak path through the cracks ***(1)***. The geometry of such cracks is not consistent; however, the defect size and its behavior are hard to define. Its behavior can be a sharp-edge orifice, a very short microchannel, or a combination of both. Updated laser-drilling technology can increase the time interval between laser shots to allow heat dissipation, effectively eliminating glass "cracking." These defects are essentially wormholes that have some consistent diameters across the entire glass wall thickness (i.e., hole-like defects). Laser-drilling manufacturers can correlate the defect "equivalent diameter" using air flow and empirical air flow calibration curves generated from known sharp-edged metallic orifices. Defect sizes less than 10 µm in glass products may show larger variability ***(1)***. The size of laser-drilled defects in glass can be estimated to "equivalent micropipette" size using calibration curves. Calibration curves should not be extrapolated beyond their ranges. When dealing with laser-drilled holes, obtaining a verification of the flow rate prior to use is recommended to prevent false negative or inaccurate results.

Figure 3.1.3-1(a) shows the cross-section a laser-drilled hole in the XY plane, which is approximately in the center of the hole. **Figure 3.1.3-1(b)** shows an image in the YZ plane, indicated by the vertical line in **Figure 3.1.3-1(a),** that shows cracks of the glass wall. **Figure 3.1.3-1(c)** shows an optical microscope image of the inner surface of a container, and **Figure 3.1.3-1(d)** shows the scanning electron microscope (SEM) image of the glass container. The star-patterned fracture passes through the wall; the crack is roughly 1 µm at its widest point.

Laser-drilled holes are often the most effective and reliable method for simulating positive controls in flexible packaging. They may be the preferred way to introduce engineered defects in such flexible containers as single-use systems (e.g., for bulk storage) or IV bags. Though, when drilling in thin-walled containers, the laser penetrates through the wall to generate a tapered hole with a throat. Another concern is that multiple small holes and channels may add up to a nominal large leak rate.

Given the flexible nature of the packaging material and the stress conditions of some CCIT methods (e.g., pressure), the hole size during and after CCIT may be different from the nominal size of the defect prior to CCI testing. Hence, variation in test results may be observed due to defect size being impacted during the test cycle.

An alternative to laser-drilling directly into a container is to use stainless-steel discs. These can be subjected to laser-drilling to create defects as small as 0.5 µm and, generally, are more consistent defects to use in equipment.

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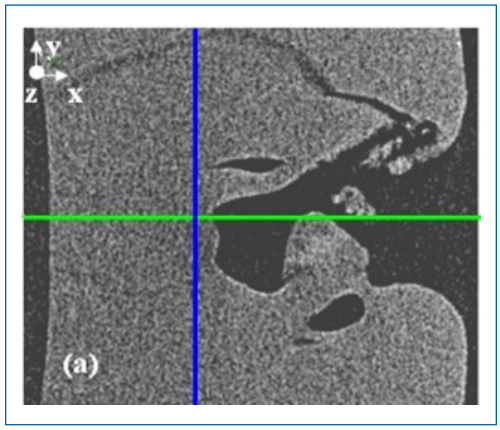

Figure 3.1.3-1(a) Cross-section of a Laser-drilled Hole in the XY Plane

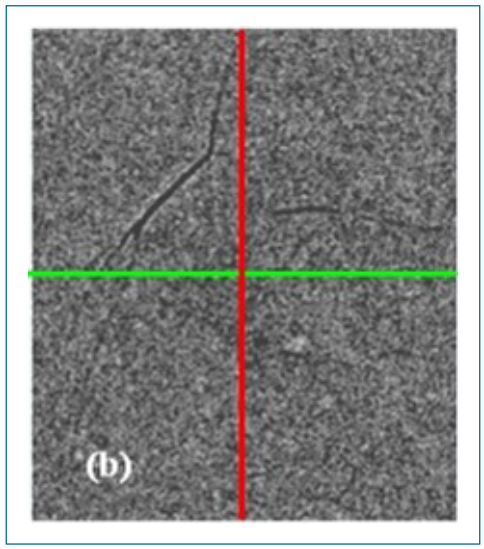

Figure 3.1.3-1(b) Image in the YZ Plane

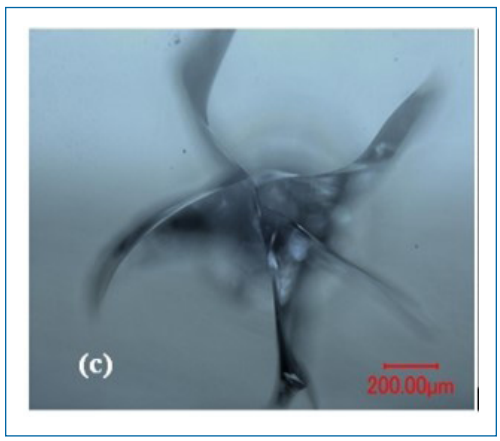

Figure 3.1.3-1(c) Optical Microscope Image of the Inner Surface of a Container

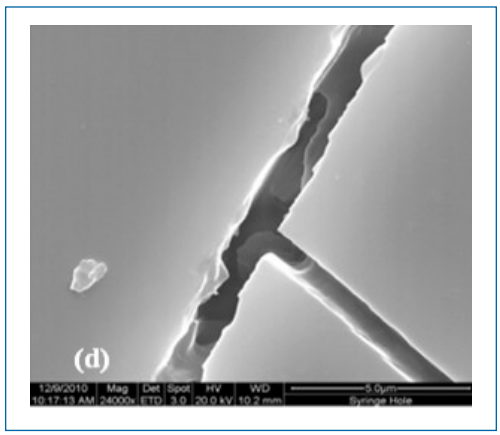

Figure 3.1.3-1(d) SEM Image of the Inner Surface of a Container

Note: X-ray *µCT* was used for Figures (a), (b) and (c)

Figure 3.1.3-1 Laser-Drilled Defect Through a Glass Vial

3.1.4 Calibrated Leaks

Calibrated leaks are designed to yield a given leakage rate calibrated against independent, traceable calibration standards. A calibrated leak is typically an independent component that is introduced into the container barrier or into the test system volume to introduce a certified known flow rate into the test system during a challenge leak test. These calibrated leaks are engineered inside a robust metallic housing and are protected from plugging via filter or clean-supplied gas. Generating a calibrated leak to mimic the headspace of sterile containers (a space with a limited air supply) is difficult because calibrated leaks need an infinite gas supply. The inlet to the calibrated leak is normally open to the gas supply source, while the outlet of the calibrated leak is connected to the equipment test circuit or test chamber via a valve. When this valve is open, a known amount of gas leakage is injected either into the test circuit (in vacuum-test condition) or out of the test circuit (in pressure-test condition). When

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Table 3.1.3-1 Summary of Common Positive Controls or Engineered Defects

Type	Possible Smallest Defect Size Available (µ)*	Advantages	Disadvantages
Laser Drilling[1]	2 µ ± 0.5 µ	Most resembles natural defects in pharmaceutical containers (cracks, pin holes). Laser-drilled holes are often a reliable method for simulating positive controls in flexible packaging	Higher cost Hole size needs to be calibrated to a leak rate Actual hole size on container may differ from nominal hole size or the hole diameter confirmation by microscopic measurement
Glass Micropipettes[2]	0.1 µ	Small defect size available Clear fluid path	Fragile Difficult to ensure a complete seal between micropipette perimeter and package wall Broken tips may not be easily detected Need to prime micropipette with media if using microbial ingress or dye ingress
Direct Insertion of Wire[3]	10 µ	Easy and inexpensive approach to creating defects that simulate channel defects Can be used for test method feasibility studies to explore method's detection range	May display gas, liquid, or microbial leakage dynamics that are very different from real-world defects
Direct Insertion of Wire/Needle/ Microfilament[4]	10 µ	Easy and inexpensive approach to creating defects that simulate channel defects Can be used for test method feasibility studies to explore method's detection range	May display gas, liquid, or microbial leakage dynamics that are very different from actual real-world defects

* *Defect sizes stated here are for information purposes only and are intended to help make decisions on appropriate positive controls that can be used with the CCS. Direct contact with vendors is recommended to get current information on the smallest defect size that can be reliably created in a flexible pharmaceutical container.*

[1] Lenox Laser Drilling Services; [2] World Precision Instruments; [3] Polymicro Technologies; [4] Artificial Leaks in Container Closure Integrity Testing: Nonlinear Finite Element Simulation of Aperture Size Original by a Copper Wire Sandwiched between the Stopper and the Glass Vial ***(5)***.

running a test with a "master container," the calibrated leak simulates a defect that is consistent and robust, which can be used to challenge and verify the CCIT equipment performance. In this condition, the calibrated leak represents a positive control sample for the CCIT equipment to detect that leak.

3.1.5 Additional Considerations for Controls

The MALL used to define the CCI level of sterile products varies across the industry. This MALL is used to prepare positive controls with simulated defects.

System suitability can be performed on each instrument prior to sample analysis using calibrated standards or other means of instrument verification. If a deterministic method is validated and instruments are calibrated, the use of positive controls to establish sensitivity may no longer be required for routine testing. The use of

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controls during validation of a deterministic method demonstrates that leaks will be detected by the method, so they may not be needed with routine testing, as long as the system is in a calibrated and maintained state. For probabilistic methods, system suitability may need to be performed along with sample analysis.

Section 6.5. provides an example of a study using positive controls, see **Section 7.0** for additional resources.

3.2 Blockage of Leak Paths

Biopharmaceutical products are formulated with a wide variety of ingredients, such as proteins, nucleic acids, or living microorganisms and buffer solution. Some of these ingredients can present challenges to CCIT. For example, some CCI physical testing methods have difficulty detecting defects in samples formulated with proteins. A study was performed to compare CCIT between a highly concentrated protein solution (120 mg/mL) and a surrogate solution with the same viscosity material property ***(6)***. The surrogate solution could flow out easily through defects, but the highly concentrated protein solution did not flow out. One hypothesis suggests that highly concentrated large molecules in solution are agglomerated around defect channels, completely or partially blocking the channels. The result was a solution that had difficulty flowing out through the channels. This supports the idea that proteins may play an important role in clogging the leak path.

Other types of defect clogging can occur when a product solution has ingredients that can crystallize upon drying. In this case, the solution easily leaks out through the defect channel, but the leaking solution will dry over time and turn into crystallized material in the defect channel. Once the solution has crystallized, it clogs the defect channel. This situation depends on the properties of the solution and the type of defect. Defect samples can be detected if they are tested soon after they are prepared, but the detection probability decreases as sample storage time increases.

When defect samples are clogged by proteins, vacuum or tracer gas-based CCI physical tests may not be able to detect defects due to the difficulty creating pressure, mass, or gas changes through the defect channel. Theoretically, if the clogged defect channels have some residual moisture and are still electrically conductive, high-voltage leak detection (HVLD) may be able to detect those leaks. Also, solution-based testing, such as the dye ingress test, may help to dissolve the crystallized materials in the clogged area if the materials are in direct contact with the dye solution, constituting a valid CCIT method for such products. However, that dye solution can also block small channels under certain conditions, such as drying or vacuum or pressure exposure.

When considering clogging by proteins or other ingredients in liquid-filled products over time, seal property, in addition to the CCIT, should be monitored in a stability program. This should be considered in feasibility studies and when selecting the CCIT method. CCIT may not be able to detect a defect in the seal if it is blocked by the clogging. Sterility could be breached if the product is contaminated prior to the clogging. Clogging is one of the challenges in CCIT; consequently, the drug manufacturers should not rely on CCIT to ensure the sterility but should develop a CCI strategy. Manufacturers can develop different ways to resolve the clogging challenge.

The CCI testing strategy and methods should address this clogging challenge. Visual inspection and RSF are examples. For example, when the seal is visible, like the plunger in a syringe, the plunger-rib contact length can be monitored to verify that the seal contact pressure does not change. When the seal is not visible, like the stopper in a vial or the disc seal in a cartridge, the residual seal force or rubber bulge height from the rubber compression can be used to verify that the seal contact pressure is acceptable.

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4.0 Challenges with Package Design

Testing for CCI is intended to provide evidence of the use of appropriately matched components and an adequate closing process. Assurance of integrity is developed from consistency of the latter design parameters. This is simple with legacy CCS such as glass vials with stoppers or glass ampoules; however, the industry has progressed to using newer product delivery designs and different materials. These newer designs are more complex and introduce a new learning curve for developing leak detection methods. **Section 4** will offer current knowledge and perspectives regarding the impact of current package designs on package integrity and the impact on the performance and selection of CCIT.

4.1 Prefilled Syringes

Prefilled syringes and cartridges are now a common dosage form for injectables. Design has progressed from standard glass prefilled syringes to plastic alternatives as well as assembled auto-injection systems. All of these advances add complexity to better understanding integrity issues and add more specific challenges to developing an approach to CCI that fits into the lifecycle assessment.

4.1.1 Design

Syringes are sealed at three key locations, called interfaces. The plunger stopper-barrel interface is the most common to all syringe (**Fig. 4.1.1-1**). Other interfaces are the needle–needle shield interface and the barrel–needle shield interface. For more detailed information on syringe design see PDA Technical Report No. 73: Prefilled Syringe User Requirements for Biotechnology Applications *(7)*.

The seal interface (or closures) for each configuration presents challenges for package integrity and CCI testing. In addition, the syringe design, materials of construction, components, manufacturing processes, secondary packaging, and auto-injectors may also impact the integrity and testing of a prefilled syringe.

Since each type of syringe has its own type of closure system, unique considerations must be made for CCI of the barrel–closure interface. To maintain CCI, proper packaging and handling procedures should be in place to ensure that syringe closures remain properly positioned throughout the manufacture, fill-finish, and transportation of syringes.

Three common types of prefilled syringe designs, regardless of materials of construction, are:

- Staked needle (barrel–closure and the needle shield–needle interfaces)
- Luer lock
- Luer slip (or luer cone)

Figure 4.1.1-2 highlights the key interfaces for staked needles.

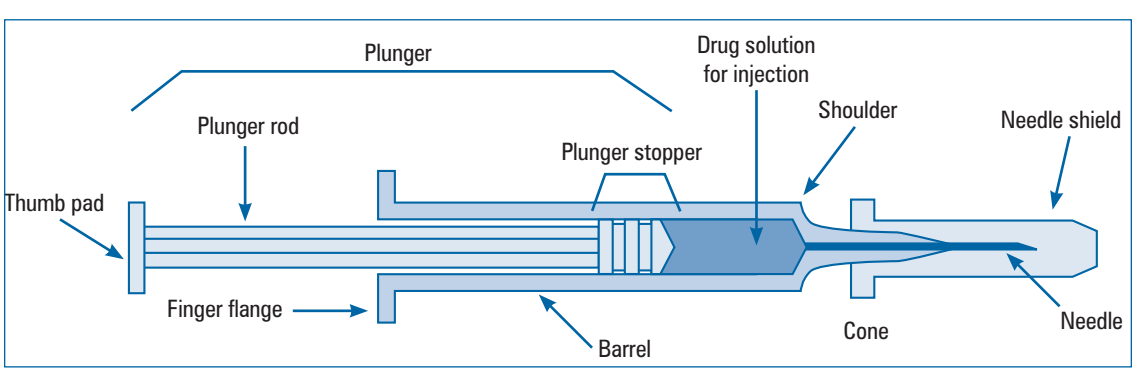

Figure 4.1.1-1 Diagram of Prefilled Syringe

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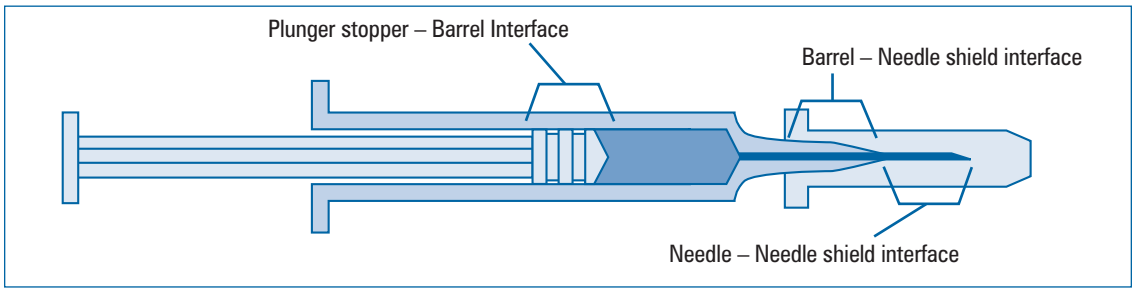

Figure. 4.1.1-2 Staked Needle Interfaces

The impact of design features on integrity and testing challenges are discussed in more detail in **Section 4.1.2.**

The barrel–closure and the needle shield–needle interfaces are included in the staked needle category (see **Fig. 4.1.1-2**). The barrel–closure interface comprises the tip of the syringe and the corresponding sealing surface of the closure. Careful consideration should be given to closure design and elastomeric formulation because different closure systems respond differently to changes in environmental conditions (e.g., heat, sterilization cycles, or CCIT method cycles). Proper interface fit between the closure system and syringe barrel is an essential requirement to ensure CCI.

The needle shield–needle interface is formed when the staked needle of the syringe is embedded into the elastomeric portion of the needle shield. The needle embedded into the elastomer is the most critical portion of the needle shield–needle interface, as it forms the closure between the drug product and the environment. The drug manufacturer is responsible for working with its suppliers to ensure that validated processes are used to properly position needle shields. The needle must be fully embedded in the needle shield, but the needle should not completely pierce through the needle shield, which can cause a defect. The drug manufacturer should then verify that additional fill-finish steps and shipping conditions do not adversely affect the needle-shield positioning or introduce needle-shield piercing.

The luer lock adapter, elastomeric tip cap, and syringe cone form the sealing surface interface for a luer lock syringe (**Figure 4.1.1-3**). Luer lock syringes with a luer lock adapter and tip cap and those comprised of a luer lock adapter, screwed rigid cap, and tip cap are the primary types. For sufficient CCI, the luer lock adapter, tip cap, and/or screwed rigid closure must be adequately positioned. During manufacturing, precautions should be taken to avoid potential silicone contamination of the luer lock adapter, which may impact the sealing integrity of the system. This may require additional line clearances to ensure residual silicone does not agglomerate on luer lock adapters.

For a luer cone (luer slip) syringe, the sealing surface interface is formed by interference between the syringe cone and the tip cap (**Figure 4.1.1-4**). Often, cone-glazing or cone-coating is used on syringes to provide surface roughness that enhances the fit of the tip cap with the syringe cone. The cone-glazing or cone-coating must be adequately distributed to ensure that the tip cap does not become dislodged during additional manufacturing steps or in transportation. Additionally, tip cap closures must be adequately placed to ensure CCI is maintained through the syringe cone–tip cap interface. Similar to the luer lock system, the presence of silicone at the tip cap of a syringe may impact sealing integrity.

For cartridge containers, CCI is maintained by a crimped combination seal (also known as a lined seal or rubber septum) on one end of the cartridge and an elastomeric plunger on the other end. The pen-injector may be a single-dose cartridge that uses a monolayer combination seal, or it may be a multidose cartridge that uses a bilayer combination seal. Cartridges may also consist of a single or dual chamber. Various options exist for dual-chambered cartridges including liquid–liquid, liquid–lyophilized powder, or liquid–powder chambers.

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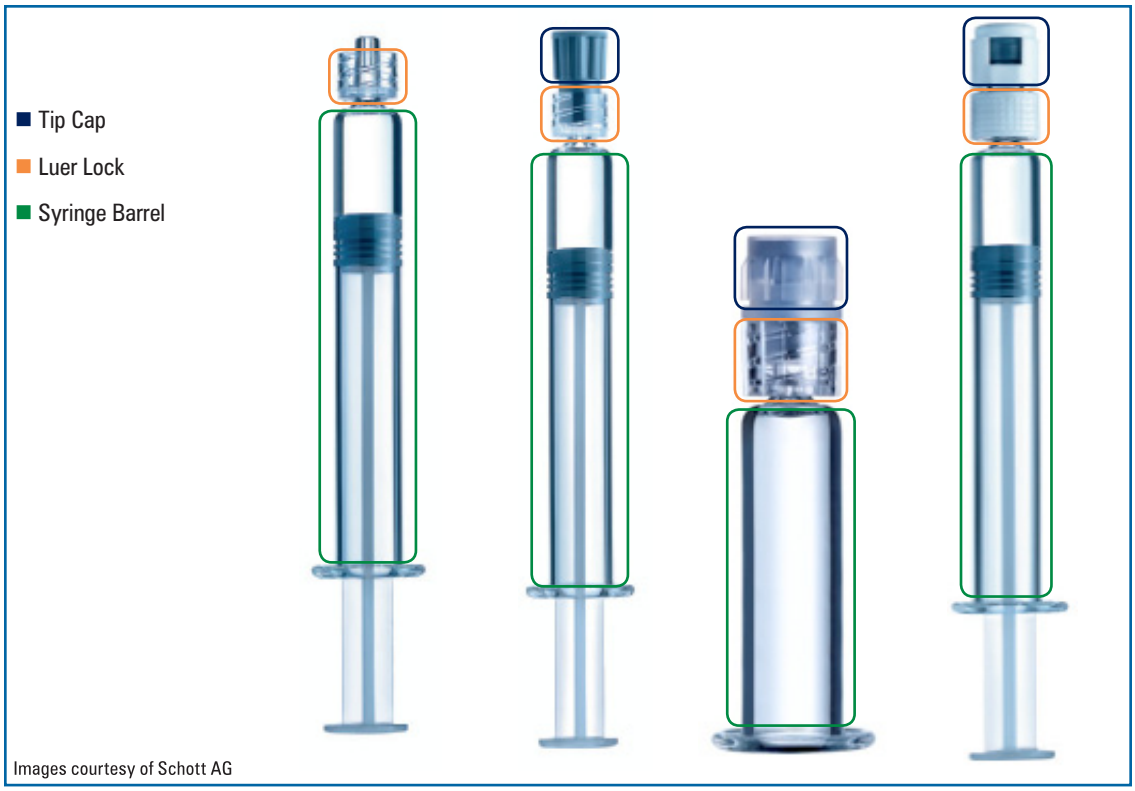

Figure 4.1.1-3 Common Luer Lock Syringes

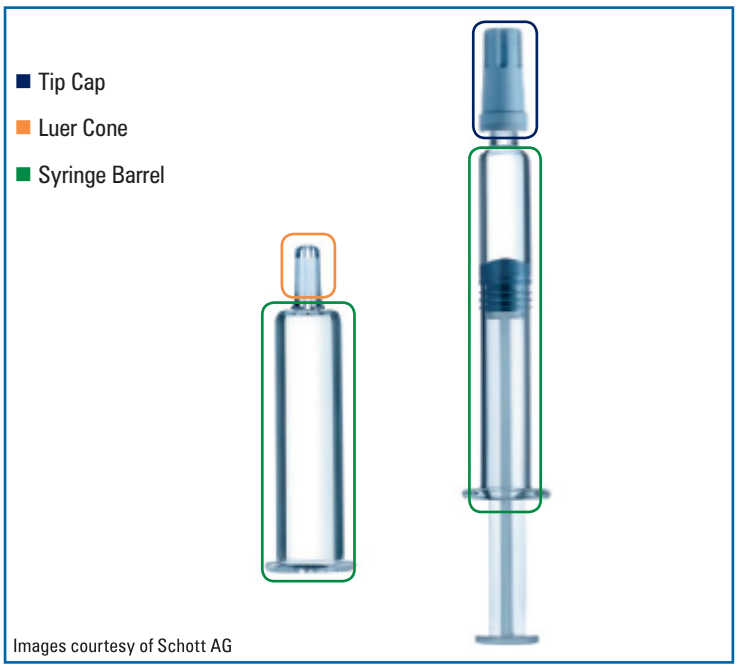

Figure 4.1.1-4 Common Luer Cone (Luer Slip) Syringes

The cartridge container must form a secure seal at the combination seal, cartridge, and cartridge plunger to ensure the system maintains CCI. For dual-chambered cartridges, specific consideration should be given to compatibility of the cartridge bypass feature (internal or external) and the cartridge plunger. Because a dual-chambered cartridge uses two elastomeric plungers for system functionality, the drug manufacturer must ensure that appropriate system-level testing is performed to maintain the integrity of the drug product in both chambers and that premature mixing does not occur.

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Combination seals perform a vital sealing function in a cartridge system that has unique requirements as compared with a syringe-based system. The combination seal must be adequately crimped to the cartridge with the appropriate compression to maintain CCI. During multiple use, the combination seal will be punctured multiple times and must maintain integrity throughout use and product shelf life ***(8)***. For multidose applications, an appropriate combination seal should be selected, including a bilayer elastomer with robust resealing properties. The drug manufacturer is expected to test the system for CCI according to the intended use of the product and regulatory test requirements, which includes testing the system with the maximum number of punctures that would engage the combination seal. Additional testing is outlined in *ISO 13926-3 Pen systems – Part 3: Seals for pen-injectors for medical use*, though the resealability test it outlines does not constitute CCI testing ***(9)***.

Elastomers and plunger stoppers provide excellent sealing properties when paired with glass and plastic containers. As with other pharmaceutical parenteral closures, halobutyl elastomers are most widely used for prefilled syringe plungers due to good drug compatibility. The elastomeric formulation must be carefully chosen and must be hard enough to provide sufficient rigidity for consistent functionality, while allowing the viscoelastic properties of the plunger material to provide closure properties.

The design of the plunger provides CCI and functionality of the system. To take advantage of the compressibility of the elastomer material, the plunger is designed to be slightly larger in circumference than the inner diameter of the chosen barrel.

The number of ribs in the design of a stopper will depend on the end use of the assembled system. *ISO 11040-5:2012 Prefilled syringes – Part 5: Plunger stoppers for injectables* recommends that plungers be designed to use all three ribs ***(10)***. In general, the rib closest to the drug is considered the primary sealing rib; the second and third ribs are meant to stabilize the seal and provide secondary CCI assurance. The design, including the placement of surface enhancements such as fluoropolymer laminate films, will also influence the CCI performance ***(11)***.

The use of lubricity coatings on the syringe barrel will influence the ability of the container system to create CCI. Syringe barrel suppliers have broad capabilities to provide lubrication for functionality; silicone oil and vapor-deposited coatings being the most commonly used. When silicone oil is sprayed on the inside of the barrel, the coating may or may not be heat-treated. The choice of treatment will play a role in the optimization of the level of application.

Lubricity coatings are applied on plungers before they are assembled into syringes to allow machinability of the components through the filling line; the coating also enhances the movement of the plunger to expel the contents of the syringe in operation. The quantity of coating applied to the surface will influence the ability of the plunger to push against the glass barrel and provide the seal.

Excess lubricity coating, while good for plunger and piston functionality and movement in use, may result in the plunger or piston moving too freely during transit, causing a breech in CCI. If the silicone oil is not heat-fixed, however, the force of the piston pushing out towards the walls of the barrel will move much of the silicone oil away from the barrel–plunger interface, helping to keep the piston in place over time. This phenomenon, though good for CCI, will make piston-initiation force difficult.

4.1.2 Challenges

Alternate materials of construction provide additional challenges with design. Prefilled syringes made of glass have long been available for use with pharmaceutical products. Polymeric (or plastic) syringes in single-use or combination formats are available as an alternate material and have some advantages over glass. Prefilled plastic or polymer syringes contain a barrel typically made from cyclic olefin polymers or copolymers. Polymer barrel materials are stronger, tougher, and more flexible, allowing for improved design and dimension control; consequently, they are used more often and, in some cases,

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seen as a potential replacement for glass syringes. The CCI requirements for prefilled syringes apply to syringes made of both glass and plastic throughout product lifecycle phases. Polymer materials and their manufacturing processes, however, present a unique set of risks for forming and maintaining seal integrity throughout the product lifecycle.

Syringes composed of polymers or copolymers present additional CCI risks. Polymeric materials are prone to undergoing physicochemical changes upon exposure to heat, oxygen, irradiation, mechanical stresses, or various combinations of such stress conditions throughout manufacturing, sterilization, and storage. These risks are related not only to the syringe barrel design, but to its molding and sterilization processes, downstream handling, packaging, and distribution processes, all of which must be identified, assessed, evaluated, and mitigated.

The supplier of plastic syringe components is responsible for establishing and validating design requirements as well as critical process parameters (CPPs) for ensuring CCI. Appropriate CCIT methods should be applied to verify that the inherent leak rate of the syringe design conforms, at a minimum, to the MALL specified in USP <1207> for preserving product sterility and drug formulation content. Such evaluations should encompass worst-case conditions in all relevant manufacturing processes, such as molding parameters, processing, and sterilization methods. Furthermore, the design and development process should include a risk assessment of the potential integrity failure modes.

Additional development studies may be required to illuminate potential surface and bulk material property changes and assess the risks of losing CCI. Ultimately, acceptance criteria should be established for the CPPs, such as dimensional specifications, relevant ISO standards, cleanliness, and freedom from defects. Adequate controls must be established throughout the polymer resin manufacturing, syringe molding, sterilization, packaging, and distribution processes to ensure the quality of components related to CCI.

The plastic syringe barrel is typically filled with product and then sealed with an elastomer plunger, which may be supplied by a different manufacturer. The marketing authorization license holders are accountable for defining the CCI requirements specific to the product, the MALL for example, and for ensuring the plastic barrel and elastomer plunger form an integral system that conforms to requirements. For semi-finished plastic syringes (product-filled and sealed), the CCI requirements must be assessed during the evolution of the product and process development, validation, and routine manufacturing phases and maintained throughout final injection device assembly (manual syringe or auto-injector devices), packaging, distribution, storage, and use. The plastic syringe design verification should consider all the components of the system, including any secondary packaging, and demonstrate that the inherent leak rate of the system conforms to the MALL required for the product. The outcome of the risk assessment should help define filling and sealing equipment requirements, CPPs, and special handling procedures. For instance, polymers are prone to surface damages by mechanical forces; therefore, special handling considerations may be needed to eliminate or mitigate damages to critical sealing surfaces.

Product stability studies should be performed to verify that the drug formulation does not significantly interact with the plastic/polymer materials in the syringe. If secondary packaging materials (e.g., labels) are expected to contact the polymer material, either directly (e.g., adhesive back of a label) or indirectly through permeation (e.g., ink on a label), those materials should be included in the studies. Appropriate CCIT methods and other relevant product attribute tests, such as visual inspection, should be included in the studies; results should be analyzed to monitor potential changes in seal quality and to verify CCI throughout the product's shelf life. Final product studies should be performed on assembled injection devices, especially when the device design applies additional mechanical stress on the polymer syringe during storage. Relevant International Conference on Harmonisation (ICH) guidelines should be considered in the study design, which may include stress conditions such as anticipated temperature excursions.

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For products sensitive to specific gases (e.g., oxygen) or when the headspace in the syringe must be preserved, in addition to gas leakage, the gas permeation rate through the plastic syringe barrel should be considered during syringe design and component selection. The total migration rate into the syringe system due to gas leakage and permeation and its impact on product stability must be assessed. Various measures can be taken to minimize gas permeation through the plastic barrel including, but not limited to, coatings both on the inside and outside of the barrels or containers, use of multilayer barrels or containers, and use of labels or secondary packaging with high barrier properties.

An appropriate CCIT method must be selected, and its applicability and key performance attributes must be characterized and demonstrated specifically for polymer syringes. Potential interfering factors associated with polymer materials, including higher gas permeation rates, out-gassing potential (especially under vacuum), and surface staining by various dye solutions, may limit method applicability, lower detection sensitivity, or cause false detections for various CCIT methodologies, helium leak detection, vacuum decay, and dye ingress testing for example. During method selection and development phases, these potential interferences must be considered, and method parameters should be tailored and optimized to minimize their impact on method performance. Use of CCI "type defects" that are common to polymer syringes, either simulated defects fabricated in the labs or real defects sampled from incoming materials, are encouraged to assist method development, and ensure method performance, (see **Section 3.1**). The final method performance must be evaluated and validated, if required, for the specific polymer syringe system and drug product.

Filling and stoppering a prefilled syringe may impact CCI, so that should be considered when testing. When using mechanical means of placing the plunger stopper, the stopper may be compressed into a tube, the tube inserted into the syringe, and a metal rod used to push the stopper out of the tube where it expands to form a compression fit inside the syringe barrel. If the plunger is laminated with a fluoropolymer coating, the compression and pushing may create wrinkles on the plunger which should be evaluated for any impact to CCI.

Vacuum stoppering is another means of stoppering the syringe; it involves creating a vacuum in the syringe, thus pulling the plunger into the syringe. Vacuum methods are less likely to create wrinkles on a fluoropolymer coating, but an incorrect vacuum setting may cause the filled liquid in the syringe to boil, creating other concerns.

The fill volume of samples should be sufficient to conduct the testing method. For example, if dye ingress testing will include spectrophotometric testing, the volume of fill must be sufficient to fill the cuvette used by the instrument. In the case of HVLD, the fill volume must be sufficient for the conducting liquid to cover any defects.

In laser-based headspace oxygen analysis, the sample fill level should allow a minimum headspace volume. The laser light should be able to pass through the headspace region of the sample without being scattered by the liquid product or by the syringe shoulder.

A filled syringe without secondary items, such as plunger rod, backstop, label, or anti-needle stick device, is referred to as a semi-finished syringe. These components items are frequently added to the syringe to aid in its functionality, identification, and safety. Once all the desired components are added, the syringe is referred to as a finished syringe.

Since the addition of these secondary items may impact different seal interface points on the syringe, the CCI of the finished syringe should be assessed. For instance, plunger rod insertion may potentially impact the seal between the plunger stopper and the inner wall of the syringe barrel, thereby deserving careful evaluation. The CCIT method should also be evaluated for any additional modification that may be required for the finished syringe as compared to the semi-finished syringe.

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Assurance of CCI of the primary package within a device assembly or needle safety system during assembly, transport, storage, and usage is critical to ensure drug sterility, stability, efficacy, potency, strength, and purity. Consideration of the impact of auto-injector (or pen) and safety system design on CCIT must be taken into account.

Due to the often-complex mechanisms of well-designed auto-injectors and pens, testing the tightness or integrity of the system is difficult. The design and content (syringe or single- or double-chamber cartridge) of auto-injectors and pens is highly variable and this variability increases the difficulty of developing a standard approach to evaluating their CCI.

4.1.3 Test Method Consideration

Decisions regarding which CCIT method to select for prefilled syringes should be based on packaged product physical properties such as density, viscosity, electrical conductivity, molecule size, and evaporation capability.

Although many methods are available for CCIT of prefilled syringe or cartridge systems, CCIT after the assembly of prefilled syringes with auto-injectors or pen-injectors creates a new set of challenges. Disposable auto-injectors are designed to avoid tampering with the primary packaging container. Consequently, using the standard battery of CCIT methods like dye ingress, HVLD, and vacuum decay may not be suitable for this purpose; instead, they must be adapted. An adaptation may involve disassembly of the device before testing, which should verify that the disassembly method does not impact integrity or accept the decreased method sensitivity that testing the intact device would allow. A similar evaluation would be warranted if an auto-injector or pen-injector is carefully cut or separated from the naked syringe.

Risk assessment of a fully assembled syringe and device and its impact on CCIT should be considered. The following factors could compromise the CCIT and may need to be evaluated as part of the design verification:

- Forces applied onto the primary container closure during final assembly, storage, and use of the device that may cause breakage or defects.
- Critical forces on the needle shield (rigid or flexible needle shield), cartridge septum, or cartridge cap.
- Critical forces on the plunger stopper (in cartridges or prefilled syringes).

Testing the actual packaged product for CCI is preferred; however, some solution properties inhibit leak detection, preventing direct testing. When the physical properties of a packaged product inhibit the use of otherwise suitable technologies, surrogate solutions may be used. For example, protein-based drug products are known for their propensity to dry in created defects; this inhibits the ability to detect those created defects by otherwise sensitive, deterministic technologies. In such cases, using a surrogate solution, such as water for injection, to demonstrate the integrity of the assembled package is widely used.

Stopper movement should be deliberately managed during CCIT. If the test methods exploit a pressure differential between the container interior and the test chamber (e.g., vacuum decay and mass extraction), stopper movement should be minimized. Stopper movement, and the resulting change in headspace volume, will reduce the pressure differential, possibly reducing the sensitivity of the method.

The application of a pressure differential to the prefilled syringe during CCIT may cause the stopper to move outward and then move back to its ordinary position once the pressure differential is removed. Stopper movement might allow entry of unwanted foreign matter; in addition, microbial contamination can be caused that could compromise the product. The adoption of technical solutions to prevent stopper movement during testing is recommended. For instance, in a vacuum decay system, one way to overcome this risk is to balance the outward stopper movement with a force equal in magnitude and opposite in direction to that of the stopper itself; this can be achieved by using a contrast pin in the test chamber.

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CCIT of auto-injectors provides challenges far beyond those presented by prefilled syringes and cartridges, and the options described in this section must be carefully considered before being implemented.

4.1.4 Additional Considerations for Secondary Packaging

A risk-based analysis should be performed to determine if any of the secondary packaging operations could impact CCI, and if CCIT should be performed to confirm that the integrity of the primary container system is maintained. If such a unit operation is identified, a CCI study should be performed on worst-case samples to demonstrate that the unit operation does not impact container integrity.

With the increasing use of self-administered prefilled syringes, secondary packaging of the filled syringe following completion of primary packaging takes place prior to distribution. Secondary packaging typically consists of (in order) plunger rod placement, finger flange placement, labeling, tray placement, and carton insertion. Depending on the dosage, additional design steps may be required. Secondary packaging of a prefilled syringe can be performed manually by a technician, by a fully automated packaging line, or by a combination of manual and automated packaging.

Plunger rod placement is the process by which a plunger rod is inserted into the elastomeric stopper. The most commonly used method is screwing the plunger rod into the back of the threaded stopper. The stopper serves as the primary barrier that maintains container integrity. Consequently, the process of inserting the plunger rod should be evaluated to determine if CCIT should be performed on the syringe after the plunger rod has been inserted to confirm that integrity has been maintained.

Finger flange placement is the process of placing a finger flange on the syringe flange. Typically, plastic syringe flanges are clipped onto the glass flange. The main risk this operation poses to CCI is fracture of the syringe. To ensure that no broken or cracked syringes enter the supply chain, a 100% visual inspection must be performed. Given that the glass syringe is the primary container, an evaluation should be performed to determine if CCIT is required.

Tray placement and carton insertion is the process by which a prefilled syringe is placed into a tray, typically a corrugated or thermoform tray, and packaged in a carton. The tray may contain one syringe or multiple syringes, depending on the packaging design. The tray is then placed inside a carton for distribution. The operation of placing the syringe in the tray, should be evaluated to determine if CCIT is required on the packaged syringe.

4.2 Single-Use Systems — Flexible Bulk Containers

The term "flexible bulk container" refers to the application of single-use flexible technologies, commonly referred to as bioprocess containers, bioprocess bags, or single-use systems (SUS), for the manufacturing process of bulk drug substance and bulk drug product. In general, these bags are available in "2-D" shapes, scaled from roughly 20 mL to 50 L, and in "3-D" shapes scaled from roughly 50 L to 3000 L in size (**Figure 4.2-1**). Historically, the use for bulk containers has been limited to less than 1000 L. With their broad range of sizes, complexity of design, and polymer material characteristics, flexible bulk containers present unique challenges for establishing inherent SUS integrity. The terms "flexible bulk container" and "single-use system" are used synonymously throughout **Section 4.2**.

4.2.1 Design

For products not terminally sterilized, the sterility of the final drug product depends on the sterility of the bulk; therefore, the integrity of the bulk container system is crucial. An integrity assessment may even be required to support a sterile claim on the bulk container for terminally sterilized products.

The critical leak size of concern for flexible containers typically relates to microbial ingress, although liquid leaks can be a critical risk to some products, for example, environmental and operator safety

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Figure 4.2-1 Mobius® 2-D Bag Assemblies from Merck Millipore and Flexsafe® 3-D Bag Assembly

related to toxicity. Gas permeation could also be a factor for product quality inside the SUS. Barrier properties to gas exchange are established during the development of the laminate material, and the protection offered may need to be assessed during the selection stage. For establishing the integrity of the SUS, the MALL must be established based on the use conditions. While SUS are out of the scope of USP <1207>, a lifecycle approach can be applied to CCI of these systems as well. **Section 6.4** discusses the lifecycle approach for bulk containers in more detail. How to evaluate the lifecycle and apply quality risk management (QRM) principles related to the SUS is described in *ASTM WK64337: New Guide for Integrity Assurance and Testing of Single-Use Systems* ***(12)***.

4.2.2 Challenges

During routine production, transportation, and handling, various scenarios and events may arise that can affect the integrity of the SUS. Qualification and a holistic approach should be used to ensure product quality as well as ensure patient and operator safety.

A good understanding of the failure mode along the product lifecycle and the result of a risk analysis may lead to implementation of nondestructive integrity testing. This testing could be performed at the supplier's manufacturing site or at the user's site, depending on the risks to be mitigated.

One key challenge for integrity testing on bulk containers is defining the critical leak size, or MALL, that the system must satisfy. Suppliers of bulk containers own the design of the whole system and therefore should be expected to qualify the containers against such a requirement. Existing literature does not support a universal maximum defect size for flexible bulk containers to determine the MALL for any specified test method. However, a maximum defect size for common use conditions, like the transport of filled bulk containers, must be determined in order to select a corresponding test method to evaluate whether the leak is present or not. The key step to determining the MALL is to measure it during the intended use condition.

4.2.3 Test Method Considerations

Microbial challenge testing is a well-known approach for identifying the MALL if prevention of microbial ingress is the critical container quality attribute. Two types of microbial integrity test methods are commonly used for primary packaging: testing by immersion and testing by aerosol. Immersion testing is generally used to assess microbial integrity, more specifically, for the storage component ***(1)***. Under normal use conditions of flexible bulk containers, aerosol challenges may be more representative; even though, historically, immersion testing has been considered both more stringent and more statistically reliable.

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Recent studies performed on rigid microtubes and flexible packaging material have recently demonstrated aerosol testing for microbial ingress to be equally as stringent as the immersion method ***(13)***. Taking into consideration the representative pressure conditions occurring during typical use for bulk containers, such as large-volume storage and shipping, a microbial challenge test by aerosol can be as stringent as immersion, showing a MALL of 2 µm no matter which method is used. With aerosol testing, the ability to easily test a greater number of samples is an advantage in reducing the impact of the probabilistic nature of the test. One option to perform this type of testing is described in the proposed standard, *ASTM Proposed Standard Test Method for Microbial Ingress Testing on Single-Use Systems* ***(14)***.

Regardless of the method, test conditions should be defined and reliably validated to simulate the expected, or even worst-case, conditions that enhance the risk of loss of integrity. Due to the flexible nature of these container systems, the variation of atmospheric pressure associated with elevation changes in air transport is not expected to increase the risk of integrity loss, but it should be verified that it will not impact the system. Higher occurrences for defects related to transportation processes, due to vibration or shocks, and to transient pressure increases above the normal static pressure from the bulk product should be considered.

Once the MALL for maintaining the critical container quality attribute of the container is determined, with targeted detection limits that correlate to the MALL, appropriate physical test methods can be developed and implemented. Both probabilistic and deterministic physical methods are available for testing bulk containers; the industry is encouraged to evaluate deterministic methods.

Ideally, the deterministic test method would have an LoD at or below the MALL. Especially for large-volume SUS and complex designs; this method is not always as sensitive as the MALL. However, it may be justified if the test method can be validated to detect process-related risks.

Suppliers of SUS have successfully demonstrated that implementation of 100% release testing correlated to the MALL is possible. Helium leak detection applied to bulk containers, which can be destructive or nondestructive depending on availability of access points, is the most sensitive technology available. Its complex nature and relatively high investment cost suggest this technology would be used more appropriately by the supplier for release testing rather than by user. Specific challenges and solutions to testing large SUS with helium are described more fully in **Section 5.0**.

In general, an assessment should be made to determine whether a point-of-use test (at user's site) will add risk to the integrity of the system. A point-of-use leak test could be replaced by transport validation and visual control for holes or other rationales. If a point-of-use leak test is considered, there are options. For example, pressure drop technology is less complex, less expensive, easier to operate, and sensitive to detecting leaks of concern to check for final integrity directly before use. Larger defects than the MALL are more likely to occur during normal operations. **Table 4.2.3 -1** compares various physical features of helium and pressure drop technologies used for SUS integrity testing.

Table 4.2.3-1 Comparison of Physical SUS Integrity Testing Methods

Feature	Helium Technology	Pressure Drop Technology
Sensitivity	≥ 2 µm	≥ 10 µm
Environmental Effects	Low	Medium (Temperature)
Volume Impact	Low to Medium	Medium to High
Material Impact	Medium to High	Low to Medium
Handling	Complex	Simple
Test Time	Low to Medium	Medium to High
Maintenance	Complex	Simple
Investment Cost	High	Low

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In general, no single test method can sufficiently assess all criteria throughout a product's entire lifecycle. Four major obstacles must be identified and resolved in order to develop a reliable 100% (all units) release-testing system:

- Size of leaks that occur in real life.
- Maximum leak size that poses no risk to microbial ingress (MALL).
- Minimum leak size that can be rapidly and reliably detected by the inspection system (LoD).
- Selection and creation of model defects for qualification of the test method.

To successfully validate test methods, both positive and negative controls must be included to demonstrate the ability of the test to detect defective containers containing an artificial leak.

4.2.4 Additional Considerations for Flexible Bulk Packaging Systems

Flexible bulk packaging systems are increasingly being used for drug substance, drug product, and fill-finish applications in the biotech industry. Sound, scientific test methods are needed to ensure their quality throughout the product lifecycle. The current model to ensure integrity is based on a quality by design (QbD) and QRM approach that should be applied by suppliers and users. Suppliers must possess a strong understanding of the design and qualification of a product for its intended use. Likewise, users must ensure their handling and other processes are compatible with the design of the SUS. Both suppliers and users should consider the method and frequency of testing as part of risk mitigation.

New technologies will continue to develop to meet the market needs, such as faster inspection times and greater sensitivity. Regardless of the capability of any single test method, it must always fit into an overall approach to quality and risk mitigation through each step of the supply chain. Ongoing research on MALLs and the comparison of real-world to model defects is required to enhance understanding and increase integrity assurance for flexible bulk container systems.

4.3 IV Bags — Flexible Finished Pharmaceutical Containers

An IV bag is a pharmaceutical product container made of flexible material, closed at the bottom and along the sides by sealing; the top of the container may be closed by fusion to IV tubing or a medication port, depending on the intended use. Because it comes in direct contact with the pharmaceutical product or formulation, IV bags are made of materials that will not alter the quality (efficacy or stability) of the product or formulation. Some of the information in this section may be relevant to considerations for bulk containers as well.

4.3.1 Design

Flexible packages often undergo temporary shape changes caused by forces within the environment, such as damages incurred during distribution and storage. The magnitude of forces sufficient to cause loss of package sterility by converting a defect into a leak must be considered when trying to determine an appropriate CCIT method that can reliably detect the leak of concern.

Imposed pressures capable of initiating liquid or gas flow through a defect provide the necessary conditions for ingress or egress to occur and, subsequently, result in a loss of sterility or quality of the product. For a specific product package system, an understanding of product's liquid properties, such as surface tension and viscosity, is useful to determine the critical leak for that system.

4.3.2 Challenges

When liquid-filled IV bags are tested for CCI, any leak from the container can pose a challenge if the product leaks and contaminates the test system (equipment and/or test chamber). The residual liquid or vapor that leaks into a test chamber from a defect in the container must be removed and the test system must be cleaned before testing can resume. This activity can affect the throughput of the test method. These challenges can also be relevant to other container closure systems.

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A defect in direct contact with air may be detected at a faster rate than a defect in contact with a liquid or product solution, and the location of the defect can have a significant impact. For a liquid-filled container, surface tension of the liquid or product must be overcome for the defect to turn into a detectable leak. The product solution of some IV bags can also dry and crystallize, potentially clogging micron-size defects and making leak detection a challenge.

The magnitude of forces sufficient to cause loss of package integrity by transforming a defect into a true leak must be considered. The role of temperature, imposed pressure, vacuum, impact, and vibration on compromising the integrity of an IV bag must be evaluated. Design studies should include testing of samples exposed to worst-case scenarios, for example:

- Stress conditions during the sterilization process (using maximum time and temperature for moist heat sterilization).
- Worst-case conditions seen during handling and storage of these containers.

Ensuring that the worst-case conditions have been accounted for during leak detection test method development is a challenge. Pinholes, channel leaks, and improper sealing all present challenges for leak detection in IV bags given that each leak can have a different geometry and pathway (direction of flow through the container). No one method can detect every different type of defect. Hence, the CCIT tool kit should encompass a variety of methods that can be employed to detect defect types critical to maintaining product quality or sterility.

Positive controls (or engineered defects) for IV bags can pose a challenge when considering leak detection through micron-sized holes. When creating positive controls, leakage dynamics as a function of different defect types and materials of construction should be kept in mind (see **Section 3.1**).

The escape of residual volatile gases and vapors from the outside surface of IV bags when they are exposed to vacuum conditions may affect leak test results by giving false positives during testing (e.g., with vacuum decay or mass extraction test methods). With such methods, the use of special fixtures or the inclusion of equilibration time before the actual test time may be necessary to limit such effects and isolate the area undergoing testing.

4.3.3 Test Method Considerations

The condition of the test specimens can have a significant impact on the test results; for example, the temperature of flex bag, time between sterilization and testing, humidity, and many other factors can influence test results. Moisture may be present on the surface of liquid-filled IV bags, especially after moist heat sterilization, which may make it difficult to detect leaks with certain technologies. Certain IV bags may not be suitable for leak detection methods that employ high vacuum or high pressure due to the potential for bursting or dislodging the container closure components.

Different methods have been considered and attempted for IV bags. Deterministic method options are HVLD, vacuum decay, mass extraction, force decay, and laser headspace analysis (tunable laser diode spectroscopy). Probabilistic method options are dye or microbial ingress, and internal pressure. An appropriate method can be selected following the guidance in USP <1207> for probabilistic or deterministic types of methods. Instrumentation should be able to, or be modified to, hold an IV bag.

Certain CCIT methods, like HVLD, detect leaks in the walls of nonporous, rigid, or flexible packaging containing liquid product. Hence, this method is not compatible with dry pharmaceutical drug products or product packages with liquid on the exterior of the package (e.g., liquid loads immediately removed from a sterilizer).

Some films that are used in the manufacture of IV bags are readily permeable to inert gases like helium, which makes application of a helium leak test method a challenge for that specific type of CCS.

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Natural or real-world CCI defects are generally larger in size than engineered or manufactured defects and represent all possible CCI failure modes. Natural CCI defects may not be pristine holes but rather long, complex, irregular channels and tortuous pathways. Where applicable, test method performance may be further supplemented using these CCI defects. Although difficult to acquire, such defects may be obtained from rejected or returned samples.

4.3.4 Additional Considerations for IV Bags

Drug products can interact with defects in IV bags in ways that can interfere with the leak detection capability of the specific CCIT method. The possible impact to product stability (e.g., HVLD or short-wave infrared imaging) should be considered in the application of certain CCIT methods.

Inherent package integrity is typically established during product-package development and manufacturing qualification. The permeation rate of gaseous molecules through the material of an IV bag must be defined and differentiated from a true leak through a micron-size defect that is critical to product quality. Some gaseous permeation may be acceptable; leakage is not.

4.4 Cryogenic Conditions

Many sterile products are stored at cold (2 °C to 8 °C), freezing (-20 °C), and ultra-cold (-80 °C to -196 °C) temperatures to limit formulation degradation and extend product shelf life. Several considerations should be addressed regarding cryogenic storage and package integrity:

- Risks to package after fill and freeze.
- Risks to large-scale bags after fill and freeze.
- Loss of integrity in frozen state due to material differences at low temperatures that are restored at high temperatures.
- Risk to package integrity in frozen state during transport and handling.

4.4.1 Design

Refrigeration storage temperatures in the 2 °C to 8 °C range are common for many products. Freezer storage at 20 °C is the next most common temperature for storage of sterile injectable products. Ultra-cold storage, beginning at 80 °C, is most commonly used for clinical trial samples where complete stability data may not yet be available (e.g., live viral vaccines); it is also the temperature of dry ice, commonly used for the shipment of product. The coldest storage temperatures are used for storing cellular materials, which use liquid nitrogen as the refrigerant at a temperature of -196 °C.

Observations of overpressure have been reported in a significant number of vials after storage at -80 °C. When syringes were inserted into these vials, the plungers pushed outwards and, when the syringe was removed, product sprayed from needle puncture holes ***(5)***. Syringe plunger movement after being inserted into a vial, or product spraying out of a needle hole, indicates a substantial overpressure inside the vial.

When vials exhibiting the overpressure phenomena were tested using laser headspace gas analysis methods, the results clearly showed changes in headspace conditions different from the specified manufacturing process parameters. The vials described should have, at room temperature, a headspace gas composition of air (20.9% oxygen / 79.1% nitrogen / 0.04% CO_2) and a pressure of 1 atmosphere**.** **Figures 4.4.1-1**, **4.4.1-2**, and **4.4.1-3** show examples of the changes in headspace gas composition and pressure after storage at -80 °C on dry ice. The measurements show excessive levels of CO_2 (atmospheric air has < 0.4 mbar), reduced levels of oxygen (atmospheric air has 20.9% oxygen), and elevated levels of total pressure (atmospheric air is 1000 mbar). The headspace changes (overpressure, reduced oxygen, elevated CO_2) have been attributed to a CCI failure while the product was stored at -80 °C.

The storage of injectable products at ultra-cold temperatures creates challenges for packaging scientists and engineers. Package integrity must be maintained at the lowest temperature a package will experi-

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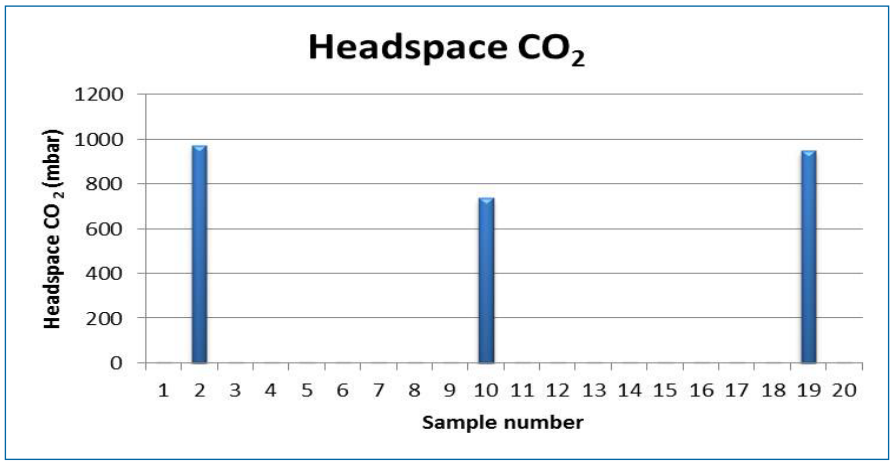

Figure 4.4.1-1 Headspace Carbon Dioxide (CO2) Pressure in a Set of 20 Vials Stored on Dry Ice at -80 °C; Three Vials Show Elevated Levels of CO_2

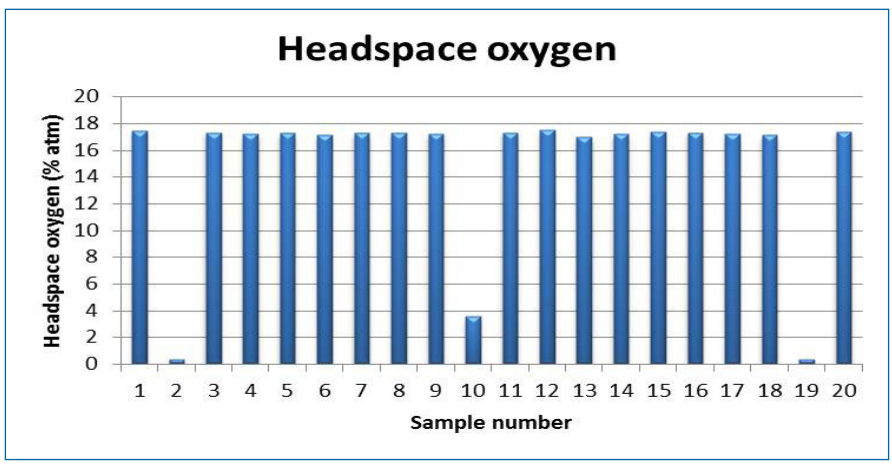

Figure 4.4.1-2 Headspace Oxygen Concentration in the Same Set of 20 Vials Stored on Dry Ice at -80 °C; Same Three Vials Show Reduced Levels of Oxygen

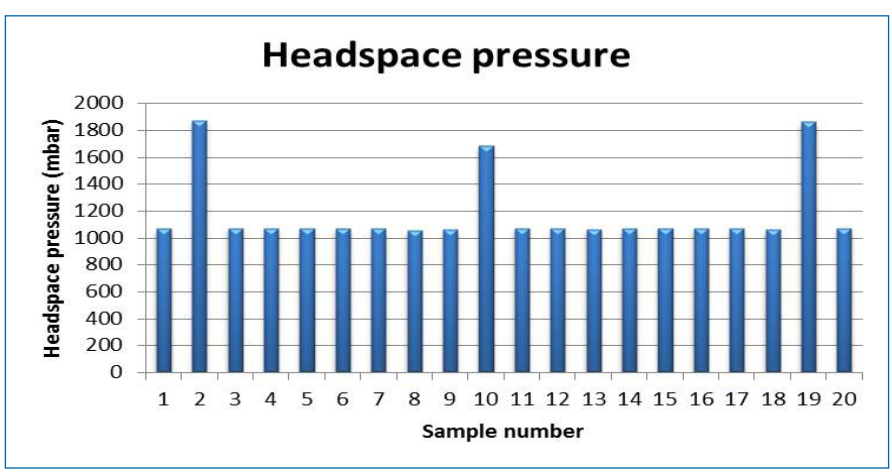

Figure 4.4.1-3 Total Headspace Pressure in the Same Set of 20 Vials Stored on Dry Ice at -80 °C; Same Three Vials Show Elevated Total Pressure

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ence over its intended shelf life. The loss of CCI at any temperature leads to sterility and stability risks to the product and patient as well as to the clinicians administering the products.

The root cause of CCI loss at cold temperatures is believed to be a result of physical changes to packaging components that occur at low temperatures. Elastomers and glass have markedly different thermal expansion coefficients, and stopper elastomers have glass transition temperatures similar to storage temperatures. The greater thermal contraction of elastomer materials, compared to glass, and the material transition below the glass transition temperature of elastomers to more plastic-like (versus rubber-like) properties, increases the possibility of a CCI failure. When CCI is compromised, nonsterile reactive gases from the surrounding environment enter the package and are trapped; once the vial warms to room temperature, the elastomer regains its elastic properties and returns to its room-temperature dimensions and fit. When high pressure, nonsterile, reactive gases (CO_2 or oxygen) are trapped inside a vial, the risk to clinicians during administration, the risk to product sterility, and the risk to the product stability increase significantly.

USP <1207> advocates a lifecycle approach to package integrity testing. CCI methods should be developed to assess package integrity for all phases of the product lifecycle—development, manufacturing, and shelf life—including assessment of CCI for a given storage temperature. For product packages that will be stored at cryogenic temperatures, CCI should be assessed as part of the package development phase.

4.4.2 Challenges

The thermal properties of parenteral packaging components, found in the common glass vial, stopper, and crimp assembly, should be given consideration.

Glass vials used to package finished sterile products are commonly made of borosilicate materials. Plastic vials used to package finished sterile products are made of polymer materials, cyclic olefin copolymer and cyclic olefin polymer. Stoppers used to package finished sterile products are primarily butyl elastomers. Each of these components have coefficients of linear thermal expansion and may also have glass transition temperatures. These parameters are influenced by changes in temperature. The change in temperature of a sterile product package is significant, from room temperature of 20 °C to ultra-cold temperature of -80 °C and, possibly, to liquid nitrogen temperatures of 196 °C. The change in the length of material due to thermal expansion can affect the fit of the components.

For example, the volume of a butyl elastomer stopper will decrease by 6% when the temperature drops from 20 °C to -20 °C, and decrease by 11% when the temperature drops from 20 °C to -70 °C (the glass transition temperature). Below the glass transition temperature, the thermal expansion-contraction rate of materials decreases, but a further decrease in temperature, to -196 °C, could result in a total stopper volume decrease of up to 20% compared to room temperature volume. In contrast, over the same temperature range (room temperature to -196 °C), borosilicate glass undergoes a volume change of less than 1%.

The greater volumetric contraction of stoppers relative to vials, and the transition of stopper material from an elastic state to a glass state below the glass transition temperature greatly increase the risk of a CCI failure.

If a sealed vial is integral at room temperature and then stored at -80 °C, the headspace pressure will decrease as the vial temperature decreases. Depending on the cold storage atmosphere (air in a freezer or CO_2 on dry ice) the increases in CO_2 or total pressure and/or the decreases in oxygen concentration can be measured.

When the temperature drops from room temperature of 20 °C to 80 °C, the stopper will contract to 88% of its original volume while the glass vial contracts to 99.5% of its original volume. Below the glass transition temperature, the stopper loses its elasticity and becomes rigid and more plastic-like.

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In addition, as the temperature drops to -80 °C, the headspace gas pressure in the vial drops to 66% of its initial room temperature value (e.g., 1 atmosphere (1000 mbar) drops to 0.66 atmospheres (660 mbar)).

The physical environment in two common cryogenic storage conditions, in terms of temperature, gas pressure, and gas composition, are an air freezer (T = -80 °C, P = 1 atmosphere, gas composition is air) and a dry ice cooler (T = -80 °C, P=1 atmosphere, gas composition is CO_2). The headspace pressure in a vial stored at -80 °C drops from 1 to 0.66 atmospheres. When a stopper–vial seal loses integrity at -80 °C, the pressure difference between the 1 atmosphere storage environment and the 0.66 atmosphere inside the vial will drive the outside gas into the vial headspace. The vial headspace pressure will equalize with the storage environment at 1 atmosphere in a time proportional to the defect size. The gas composition, when pressure equilibrium is initially reached, will be 2/3 air and 1/3 storage environment (either air or CO_2). If a leaking vial remains in cold storage once the pressure equilibrium is reached, a different physical leak mechanism, diffusion, continues to cause exchange of the nonsterile storage atmosphere with the vial headspace. Over a period of time, depending on the size of the leak, the atmosphere in the vial (both pressure and gas composition) will come to equilibrium with the storage environment.

When the vial–stopper–crimp package is removed from cryogenic storage and warmed to room temperature, the stopper will begin to expand, and the above-the-glass transition temperature will regain its elastic properties. The vial–stopper–crimp combination will reseal and trap the cold dense gas in the vial. As the temperature continues to rise in the now-resealed vial, the headspace pressure rises linearly until the package reaches thermal equilibrium with room temperature.

In this vial–stopper–crimp example, if a vial is stored at -80 °C on dry ice and loses CCI below the glass transition temperature of -70 °C, the vial will fill with 0.35 atmosphere of CO_2. Over a longer storage period, the oxygen in the vial will diffuse out and be replaced with CO_2. If the vial is then removed from dry ice storage and warmed, the stopper will regain elasticity at the glass transition temperature and the vial will reseal. The headspace pressure will increase linearly in temperatures from 70 °C to 20 °C; the headspace pressure will increase from 1 atmosphere at -70 °C to 1.4 atmosphere at 20 °C; and the gas composition will contain high concentrations of CO_2 and reduced concentrations of oxygen.

4.4.3 Test Method Considerations

Measuring changes in headspace gas composition and pressure after cryogenic storage can provide insight into whether a package has maintained CCI. A key point is that the CCI failure is temporary. As the vial warms up, the elastomer regains its elastic properties, expands, and reseals the vial. Some methods are well suited for measuring CCI failure. Laser headspace analysis is uniquely suited to study changes nondestructively in headspace gas composition and pressure to determine CCI failures. Helium leak detection and residual seal force measurement can also be used to indicate changes from cold conditions ***(15)***.

When CCI is lost in a sterile pharmaceutical vial stored at cryogenic temperatures, the gas composition and pressure in the vial headspace will change, and headspace analysis can measure those changes. Depending on the cold storage atmosphere (air in a freezer or CO_2 on dry ice), the increases in CO_2 or total pressure and/or the decreases in oxygen concentration can be measured**.**

4.4.4 Additional Considerations for Cryogenic Conditions

Several studies have been published in the *PDA Journal of Science and Technology* or presented at PDA meetings on the study of CCI failures of sterile product packages stored at cryogenic temperatures and other established sources ***(5,11,16–18)***. Some of these thermal properties of parenteral packaging components, found in the common glass vial, stopper, and crimp assembly, should be given consideration.

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5.0 Innovative Methods for Existing Technologies

The pharmaceutical industry is constantly searching for new package integrity testing technologies to support the sterility assurance of drug products. Two innovative CCIT methods—helium testing and optical emission spectroscopy—and two seal-integrity methods— ultrasound and x-ray detection—are described in **Section 5**. Though these methods may be improvements in technology, they, too, present limitations and challenges.

5.1 Helium Testing

Helium integrity testing is a well-known, deterministic, nondestructive test method that has been used for many years in several industries to test rigid containers. The technology, which has special applications, now allows the testing of flexible polymeric container systems, such as those described in **Section 4.2.**

5.1.1 Application

Helium testing is suitable for test units that are dry and can withstand at least 1 psi pressure differential once they are restrained by a special holder. This test method can be used to verify the integrity of fully-assembled 2-D and 3-D single-use bags (such as disposable bioreactor or mixing and storage bags) as well as complex manifold systems consisting of multiple 2-D or 3-D bags connected to a central tubing manifold (**Figure 5.1.1-1**). Specific components that have a natural leak rate above the determined limit, e.g., aseptic connectors with porous membranes, cannot be tested and must be excluded. Permeation leak rate of nondefective assemblies should also be considered. The complex assemblies, which have more than 100 connection points (such as barb–tube, film–film, and film–fitment joints), can be tested to a defect size of a 2 µm pinhole equivalent flow-effective diameter.

The method involves placing a finished biopharmaceutical single-use bag assembly into an enclosed test chamber, evacuating the chamber to create a vacuum, and then introducing a measured quantity of helium gas into the bag assembly being tested. A specialized pumping technique reduces the stress on the bag assembly by reducing the internal pressure of the bag along with the external (chamber) pressure. In addition, the test chamber can be equipped with restraining plates and porous spacers to mechanically support the assembly under test and avoid the masking effect of defects. The helium-filled bag assembly is then held for a fixed time while the surrounding vacuum is monitored for leaking helium using a sensitive mass spectrometer. An automatically preceded pressure-sensing gross leak check prevents the test chamber from being saturated with large quantities of helium, practically

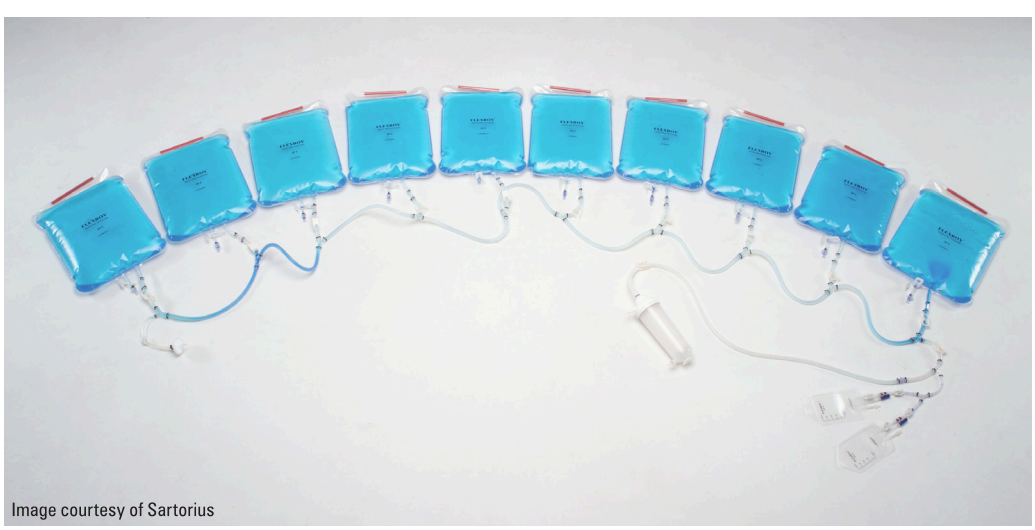
Image courtesy of Sartorius

Figure 5.1.1-1 Sample Manifold with 10 2-D Bags Attached to a Central Fluid Line with 70+ Joining Points

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eliminating the system down times due to helium pollution. This method has the advantage that it can be made semi-automatic, reducing the role of the operator.

5.1.2 Test Equipment and Parameters

The helium test apparatus includes a vacuum test chamber, pressure sensors, a vacuum pump, and a path providing the helium to escape. The chamber can optionally be equipped with restraining plates and porous spacers constraining the bag volume. The test chamber is connected via two lines: the first line evacuates the test chamber and, in the process, carries any helium molecules in the test chamber to the detector; the second line introduces helium into the bag assembly undergoing the test.

Typically, this test is performed in a vacuum chamber with low test pressures, at room temperature. The test time varies depending on the complexity and/or volume of the bag design being tested. The critical parameters impacting the test results are defect size, helium permeability of a single-use bag wall, test pressure, test temperature, bag-tubing surface area, helium injection volume, and test time (time during which helium is collected). Actual test parameters must be obtained by collecting data for defective versus nondefective bags that provide the highest difference between the two.

Helium molecules escaping through the defects are sensed by mass spectrometer and are related to a leak rate through a known defect size.

5.1.3 Leak Detection Limit

Helium testing can sense helium leak rates in the range of 10^{-8} to 10^{-4} atmosphere mbar L/s. The test sensitivity depends on the background helium leak rate of a nondefective bag assembly versus one with a defect. The greater the difference between the background leak rate of the nondefective bag assembly versus the defective bag assembly, the higher the confidence that a defect can be detected. Helium testing is capable of detecting 2 µm laser-drilled defects, as shown in **Figure 5.1.3-1** (examples of helium leak rates of a nondefective bag versus a defective bag). Just as design-specific characteristics like numbers of connections, surface area, and type of film and tubing material can highly impact the natural background leak rate of nondefective assemblies, an individual acceptance limit must be identified and proven with a statistically significant number of positive and negative controls.

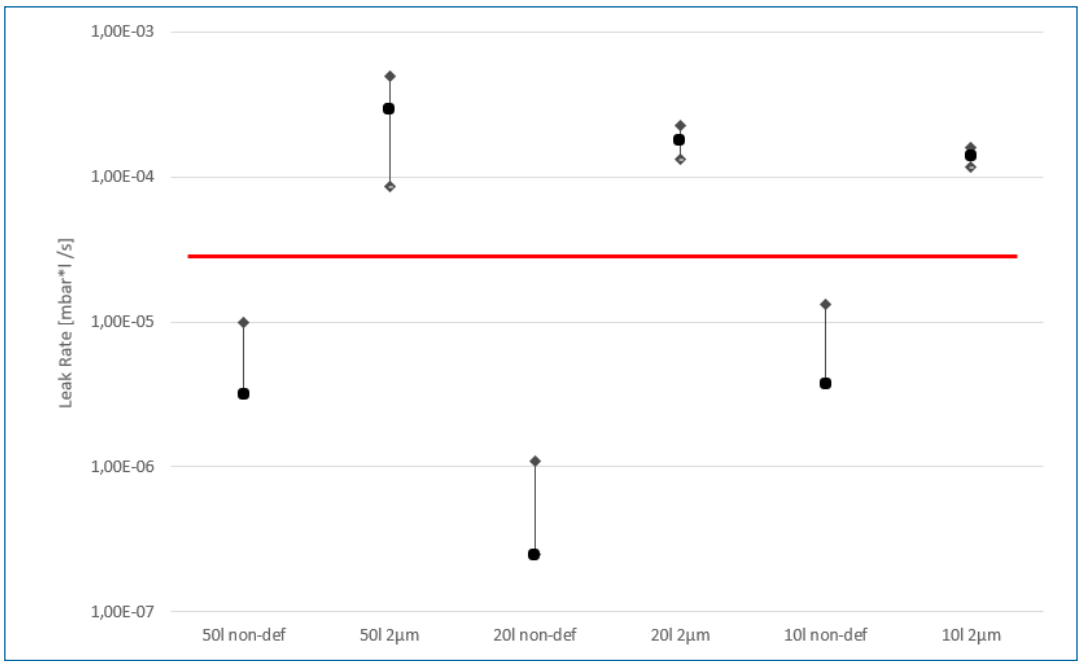

Figure 5.1.3-1 Comparison of Helium Leak Rates of a Nondefective Bag versus a Defective Bag

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5.1.4 Considerations and Challenges

Instrument performance can be verified by using standard calibrated helium leaks. A helium leak rate through the actual test sample with a known defect size versus a helium leak rate through nondefective samples helps determine the smallest detectable defect for a given test unit. Helium testing provides a quantitative leak rate. The test can detect the presence of multiple defects, but not the location of a defect.

The preferred test method uses test units with reasonable gas barrier properties. If the test unit is highly permeable to helium, then sensitivity drops significantly. The risk of false positives can be minimized by incorporating an in-line leak in the instrument to ensure that the sensor is in good working condition during each test cycle.

5.2 Optical Emission Spectroscopy

The emission spectrum of a chemical element or compound (atom or molecule) is the spectrum of frequencies of electromagnetic radiation emitted due to the atom or molecule making a transition from high energy state to a lower energy state. The energy of the emitted photon is equal to the difference between the two states. For each atom, there are many possible electron transitions, and each transition has a specific energy difference. This collection of different transitions, leading to different radiated wavelengths, makes up an emission spectrum. Each element's emission spectrum is unique; therefore, optical emission spectroscopy (OES) can be used to identify the elements in matter of unknown composition or to perform chemical analysis of substances.

OES analysis does not require any specific tracer gas to measure total leakage on sealed containers ***(18)***. Instead, the gas mixture naturally present in the container headspace of the primary container is used to perform high-sensitivity tests over a large detection range.

5.2.1 Application

The OES equipment measures the total leak rate for a specific gas from a sample under vacuum. The test method is deterministic, nondestructive, easy to use, easy to set up, and highly sensitive.

OES has been developed to measure or check the CCI of a variety of sterile CCS such as pouches, vials, prefilled syringes, and cartridges. In the case of prefilled liquid products, water leaks coming from the surface and air leaks coming from the air headspace can potentially be measured simultaneously.

The ability to analyze the molecular signature of the leaking gas helps troubleshoot the source and location of the leak (e.g., for a CO_2-filled liquid package, a higher level of CO_2 indicates headspace leakage). In addition, the molecular signature may reduce false rejects (e.g., for dry products filled with nitrogen, an unusually high level of hydrogen may suggest moisture leakage into the chamber due to chamber-seal leakage).

For sterile glass products, the low outgassing rate of the glass vials and syringes allows testing of the products in batches of several syringes, potentially increasing the throughput while maintaining the high sensitivity of the test.

5.2.2 Test Equipment and Parameters

The OES equipment includes a certified calibrated leak, which is used for the calibration. The leakage is measured in leak flow unit of mbaržL/s.

The OES sensor is connected directly to a test chamber in which the container to be analyzed is loaded. The test chamber is evacuated and, at low pressure ($< 5.10^{-2}$ mbar), the OES sensor is able to generate plasma. The excited atoms and ions in the discharge plasma generate light through numerous characteristic emission spectral lines generated by the elements in the gas mixture and captured by a camera. The wavelength and intensity (amplitude) of each emission line is analyzed. The emission intensity

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(amplitude) is dependent on the partial pressure (mol ratio) of the related element in the gas mixture, and change in intensity over time is proportional to leakage and/or outgassing (desorption of the gas from the container surface).

Testing time is a direct consequence of the desired LoD in combination with the total outgassing flow, which depends on the materials under vacuum and the relative humidity in the ambient air trapped on the container.

Special considerations and set-up parameters should apply to testing containers with high outgassing materials and dry products with small headspace applications.

The test sequence has three major steps to check the integrity of the CCS. The operator loads one or more test samples into the test chamber and starts the test.

1. **Rough Evacuation Step.** The chamber starts at barometric conditions and is evacuated down to a few mbar. At the end of this step, if the set vacuum level is not reached due to a gross leak, the test will stop and a proper test failure message will be displayed.

2. **High-Vacuum Evacuation Step.** When the pressure inside the test chamber reaches the maximum allowed pressure for the turbo pump (typically a few mbar), the OES sensor is in contact with the test chamber and pressure can be measured. It takes one or two seconds to reach steady measurement due to pressure fluctuations.

3. **Measurement Step.** The OES sensor is measuring continuously while the pressure is continuously decreasing. The test duration has been defined during the calibration of the equipment in order to achieve a signal-to-noise ratio large enough to reach a good capability. Calibration is done during initial set-up of a given product and includes a negative sample and the calibrated leak with the proper gas at its inlet.

5.2.3 Leak Detection Limit

The instrument measures the leak rate for any gas in a container (e.g., oxygen, nitrogen or even water vapor) escaping from a leaking sample under vacuum. Because the test is performed under high vacuum, the driving parameter of the OES method sensitivity is the outgassing rate of the container material in contact with or exposed to high vacuum. The smaller the leak, the longer the outgassing takes for creating a measurable pressure difference, and this rate or better the amount of gas escaping from a leak determines as a consequence the LoD for a given test time. Due to the physical constraints under those high-vacuum conditions, water exists only in vapor form, which enhances the LoD towards smaller leak sizes when the leak is filled with water because an evaporating droplet of water creates significantly more gas as if it would escape from the head space of a container. In other words, in the case of a small dry leak, compared to the same small leak with liquid behind it, due to a constant water evaporation during the measurement cycle, the amount of gas escaping through the leak is far higher and the detection limit for those holes is also shifted toward much smaller hole sizes.

The upper detection limit, the limit of how large a leak can be to become detected with a differential pressure method, depends on the size of the headspace volume compared to the amount of liquid in the container. This is the result of creating high-vacuum conditions in a test chamber, usually by evacuating the chamber with vacuum pumps over time. If a leak is very large relative to the container size and fill level during this evacuation cycle, all gas and water vapor could be evacuated before the test cycle can start. Under these conditions, the test will produce a false-positive detection, which means a leaking product can pass as an integral product because the container is already empty. Usually those conditions do not occur on all container and fill level variations, but it should be considered as a risk factor before choosing the method. If it can occur, assisting the quality check with a visual inspection method that can detect a malformed container before or empty container after such a leak detection is recommended.

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5.2.4 Considerations and Challenges

A number of factors should be considered when using OES, such as safety considerations and the design of the vacuum chamber. Local safety rules and regulations should be followed at all times, and particular attention should be paid when handling toxic drugs, especially if large leaks may be present.

Design of the vacuum chamber affects both testing and packaging. Proper chamber design is critical to contain liquid spills and minimize test volume. Tests are performed at high vacuum (under 5×10^{-2} bar-absolute), during which large-molecule drugs may plug the leak path; consequently, this method should be qualified for the specific drug being tested. In addition, flexible packages, or packages with moving parts (e.g., prefilled syringes), must have proper vacuum chamber design to constrain expansion from causing damage to the package. Flexible packages with large outgassing, such as IV bags, should be tested based on the specific application, as outgassing depends on the package material.

5.3 Airborne Ultrasound

Ultrasonic waves are high-frequency sound waves that exhibit behaviors characteristic of sound waves. When a boundary exists between two media with varying acoustic impedances, the ultrasonic waves are partially reflected and partially transmitted through one medium into the other. Additionally, refraction, absorption, and diffraction occur at various instances. Airborne ultrasound is a point-focused signal that pulsates at 500 pulses per second.

5.3.1 Application

Airborne ultrasound, a seal integrity test, is a "through transmission technology" that transmits ultrasound through a material and receives the signal on the opposite side of the material. Since, porous packaging film allows the passage of gases (such as air) while maintaining a sterile barrier, traditional vacuum decay or tracer gas leak tests cannot be used to ensure its integrity. Airborne ultrasound seal inspection allows the seals of porous packaging to be inspected online, quantitatively and nondestructively ***(19, 20)***.

Porous sterile barrier packaging is one of a variety of flexible package formats available. Creating a functional flexible barrier package typically requires bonding two films together to create a flexible package structure, then sealing it once it has been filled with the product. Each of the pouch seals is critical to the integrity of the pouch, and measuring the quality of those seals is a means of ensuring the integrity of the flexible packaging.

Visual inspection of both porous and nonporous flexible packaging assumes that all defects are visible. The quality of a seal is dependent only on physical properties; therefore, the visual appearance of a seal area does not ensure the integrity of the seal. Heat-sealing bars can leave a seal imprint on the outside of a package, while the inner seal fails to reach bonding temperature and never seals completely or has a weak peel strength.

5.3.2 Test Equipment and Parameters

The airborne ultrasound system consists of an ultrasound sensor head and a signal processing system. The sensor head is used to pulse sound at a bonded film material and measure the signal strength on the opposing side. The ultrasound signal processor gathers the ultrasound pulsing measurements and processes the measurement into quantitative summary data for evaluation.

The sound is focused on an active area relative to frequency (or wavelength) of the sound. The active area determines the resolution of the focused measurement. Airborne ultrasound is primarily suited for film materials, providing an acoustic signature of bond materials or material characteristics based on thickness, density, and material uniformity (**Figure 5.3.2-1**).

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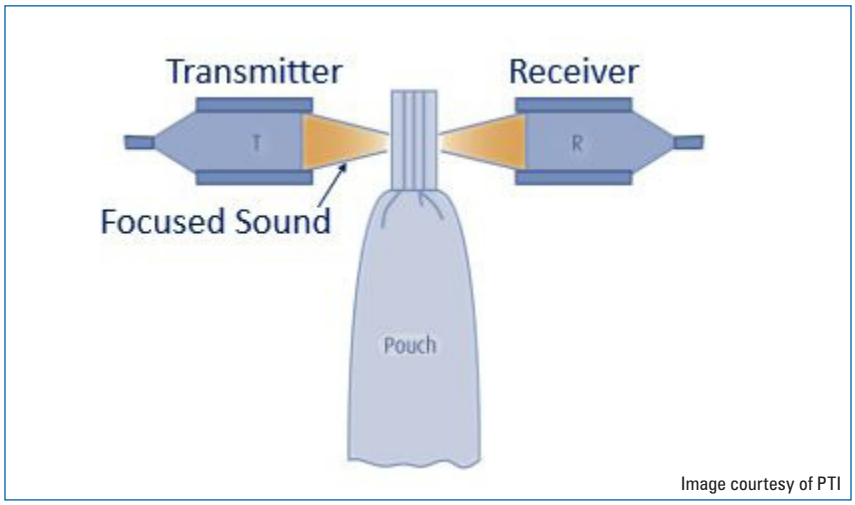

Figure 5.3.2-1 Airborne Ultrasound Through Transmission

The principle of ultrasound measurement relates to double-paned, soundproof windows. A single-pane window allows sound to propagate through the single pane and be measured on the opposing side. A double-pane window absorbs and reflects most of the sound waves, preventing the majority of sound from reaching the opposing side. Relating this principle to airborne ultrasound, the sound waves will propagate through a well-bonded seal (a single pane, **Figure 5.3.2-2**). The signal transmitted through a bonded seal is measured and used to determine seal quality. A weak seal, seal defect, or point of nonbonding will prevent the ultrasound from propagating through the seal. For multiple layers (no seal), the signal through the seal is blocked completely (like the double-paned window).

The ultrasound signal is a point-focused measurement. Test data comprises signal intensity associated with a position on the material or seal area and the decibel-level signal strength. The strength of signal value provides the measurement of seal quality, and that measurement is paired with the position within the material or seal area.

Seal defects can create additional transitions of plastic-to-air or plastic-to-foreign contaminant, causing multiple reflections and refractions. Trapped air or sufficiently large solid contamination (≥1 mm) could also cause the signal to attenuate or decrease in strength. Alternatively, the presence of cuts or holes in the seal would result in a peak in the signal strength. The ultrasonic signal can provide an acoustic signature of bonded materials or material characteristics based on density and material uniformity.

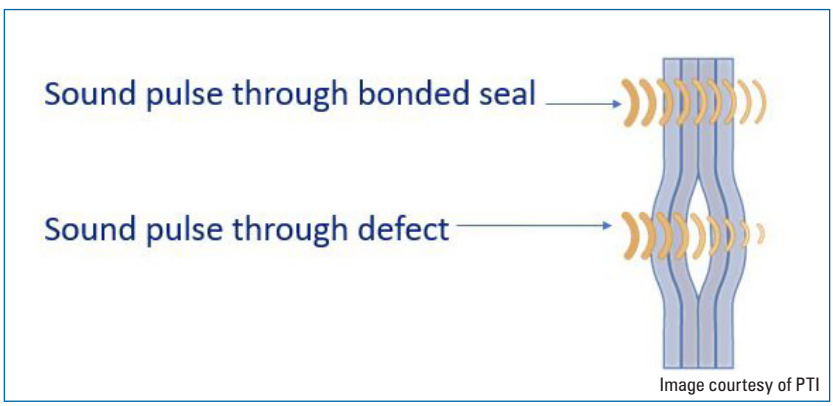

Figure 5.3.2-2 Propagation of Ultrasound Through Material Being Tested

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The measurement can be presented in a line graph format (**Figure 5.3.2-3**), or a composite scan can pair line scans to create an optoacoustic image (**Figure 5.3.2-4**). The optoacoustic image is a data matrix of ultrasonic signal strength data paired with X-Y coordinates. The data is presented in the form of a bitmap file, making the image quantitative in its base measurement. Both methods of presenting data provide quantitative and visually qualitative assessments of materials and seal quality. Quantitative measurement results characterize different properties of seals and seal materials.

The ultrasonic measurement can be performed in a variety of ways. In order to provide such measurements, the seal must pass through the inspection head. As the seal passes through, moving at a rate that allows the pulses to overlap slightly, the ultrasonic sensor emits chirps at a specified rate. The most efficient way to perform seal inspections is to scan seals directly after they are made on the production line.

While an optoacoustic image can be used to evaluate seal quality, the linear scan inspection mode is the most appropriate for discrete defect detection. When a linear scan is performed on a package seal, the system gathers the signal strength data of all the pulses applied to the seal area. The leak detection is based on processing a series of high-frequency data points and compiling the population of data into summary statistics for each scan.

Once a scan is complete, the data points are recorded, and the summary statistics are produced for the scan. The summary statistics consist of the average signal, maximum signal, minimum signal, standard deviation of the signal, and length of the scan.

A low signal value or minimum value usually indicates the presence of a defect such as contamination, inclusions, delamination, voids, or no seal. It may also indicate the test was taken beyond the seal. A very high signal or maximum value indicates the presence of a cut or gap in the seal, or that the test was taken too close to the edge of the sample.

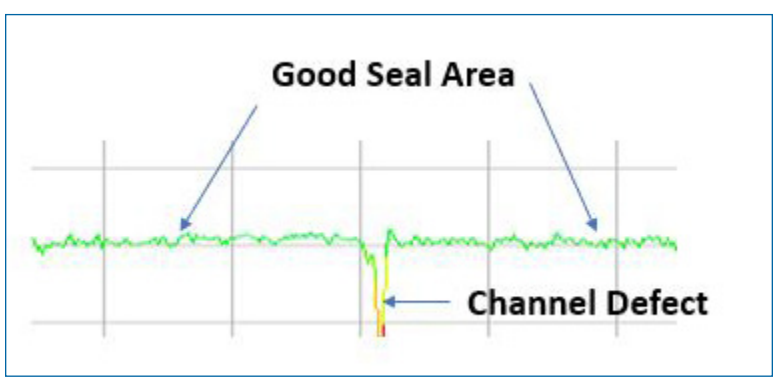

Figure 5.3.2-3 Ultrasound Single Linear Scan Graph of Channel Defect within a Pouch Seal

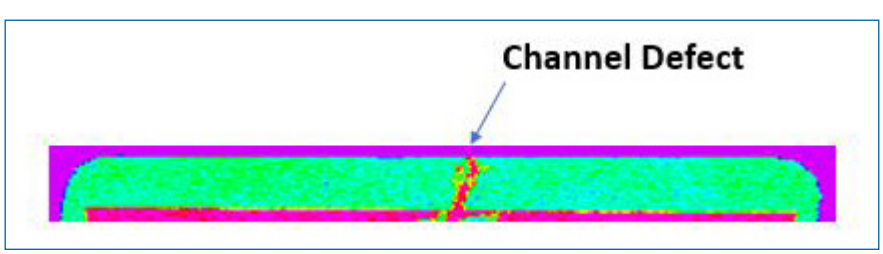

Figure 5.3.2-4 Ultrasound Composite Scan/Optoacoustic Image of Channel Defect within a Pouch Seal

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The ultrasound system uses the summary data to provide the discrete pass or fail test result. The resulting statistical summary data from each scan are then evaluated against acceptance limits for each summary statistic. If any of the statistical summary data points fall outside the acceptance limits, the test result is a "fail." If all the summary statistics fall within the limits set, the ultrasound scan is a "pass."

5.3.3 Leak Detection Limit

Defect detection using airborne ultrasound is based on the quantitative assessments of a scanned seal. As described in **Section 5.3.2**, the defect is detected by breaching a set-point limit for a summary statistic; defects can also be visually apparent in graphical form. The ability of the airborne ultrasound to detect a defect is based on a variety of factors. Defects that can be detected are typically cold seals, channel defects, wrinkles, areas of nonbonding, poor seals, material variations, seal variations, cuts, or holes. The ability to detect these defects is primarily dependent on the physics of ultrasound and the material properties.

5.3.4 Considerations and Challenges

Two additional factors that need to be evaluated before an application can be considered are the package contents and the seal geometry. With package contents, many seal defects occur due to product inclusion in the seal area. Liquids can propagate an ultrasonic signal effectively from one material to another, which means the sound would transfer through a defect if liquid were present in the defect. The presence of liquids in the seal can produce test results similar to that of a good seal.

Seal geometry can also affect seal inspection. The ultrasonic signal is a focused beam of sound, and approximately 90% of the sound remains within the target focal point. Some sound energy resides outside the focal point. If a seal is located on the edge of the material and a scan is performed on the edge of the material, sound will travel around the edge and will be measured on the receiving side. The ultrasonic sensor must be within 2 mm to 3 mm inside the pouch to produce a useful ultrasonic signal response. As the sensor nears the edge of a seal, the ultrasonic signal is impacted by sound reflecting within the seal, as well as sound passing around a seal edge to reach the receiver. These two factors limit the ability of an ultrasound to scan near the edge of a seal.

5.4 X-Ray Detection

X-ray detection may not be considered a leak test, as it neither quantitatively nor qualitatively evaluates leak rates or leakage; however, X-ray technologies support CCI evaluations in a variety of ways. Technological breakthroughs in manufacturing of less costly and more sensitive large detector arrays, as well as the increased acceptability of products exposed to X-ray radiation, have resulted in an increased potential for its use.

X-radiation (composed of X-rays) is a form of electromagnetic radiation that has a wavelength in the range of 0.01 nm to 10 nm, corresponding to frequencies in the range of 30 PHz to 30 EHz (3×10^{16} Hz to 3×10^{19} Hz) and energies in the range of 100 eV to 100 keV. X-rays are shorter in wavelength than UV rays and longer than gamma rays. X-rays can be generated by an X-ray tube, a vacuum tube that uses a high voltage to accelerate the electrons released by a hot cathode to a high velocity. The high velocity electrons collide with a metal target, the anode, creating the X-rays.

5.4.1 Application

X-ray is applied to package systems to evaluate physical attributes of the test system that may not be visible to the naked eye or even to some visual inspection systems. For this reason, X-ray detection, in its current state, is best used as a supporting technology rather than a leak test, as a supplemental means of characterizing a package system with respect to CCI. Physical attributes may include defects conducive to leakage, such as poor stopper fit or cracks in a glass vial or syringe, but not necessarily the propensity for gas or liquid to leak, or flow, through such defects.

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5.4.2 Test Equipment and Parameters

For pharmaceutical CCI applications, an X-ray image can be produced in a number of ways. The first is a line scan technique (**Figure 5.4.2-1)**, where the object to be inspected is slowly moved through a plane of X-ray radiation and the picture is created in a linear X-ray-sensitive diode array. This technique is called fan or line-beam X-ray scanning. The advantage is high resolution, although from a very low speed (because the picture needs to be composed line by line), and the scan speed defines the resolution.

A variation of this scanning technique known as X-ray tomography, or CT (computerized tomography) scan, employs the use of multiple X-ray scans at different angles to reconstruct a 3-D volume rendering of the external and, importantly, internal geometries of the sample. This is more commonly performed through the use of a cone-beam X-ray scanner. Cone-beam scanning is similar in nature to line-beam X-ray scanning but differs in that the sample is placed on a rotating stage from which a cone of X-rays is emitted. CT scans have the advantage of high resolution and complete 3-D rendering, enabling the operator to evaluate a single package from an array of angles, increasing the analytical value of a single scan. However, a single scan may span minutes, hours, or even days depending on instrument capability and resolution (scan speed).

Another technique for X-ray analysis is the area X-ray camera, which works much like a simple photography camera. Here, the object will be penetrated by the radiation, attenuated by the object, and the result captured as a gray-scale image on a large X-ray-sensitive diode array (See **Figure 5.4.2-2**). This scanning process produces 2-D as well as 3-D pictures, but 3-D is widely used in material research.

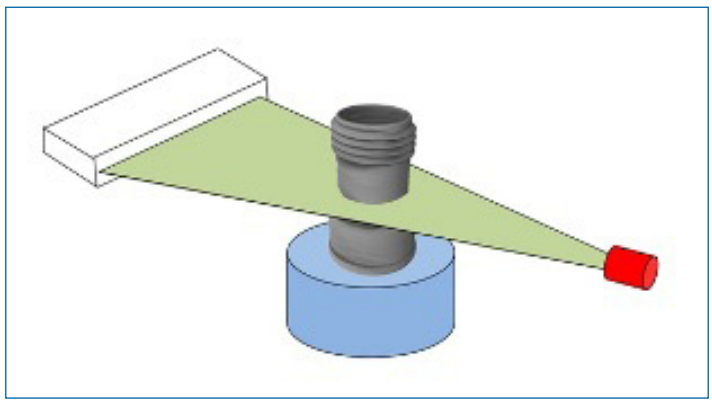

Figure 5.4.2-1 Principle of the Line Scan Imaging Technique

Figure 5.4.2-2 Principle of the Area Scan Imaging Technique

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The area imaging technique is fast and reliable because the pictures are taken during a complete standstill of sample and beam. With this technology, a picture can be produced and evaluated within 100 ms or faster compared to scanning technologies.

In addition to the detector and technique, the X-ray source itself affects crucial result properties. These properties are defined by the application field. In short, there are two types of X-ray tubes. The first is the closed-tube design, which produces radiation with a focus between 800 µm and 1500 µm. The most useful application field, as it pertains to CCI evaluation, is referred to as the microfocus tube design, which has a focus between 1 µm and 5 µm. In general, the focus defines the spatial resolution of an imaging system or the smallest object, which can be made visible in a resulting picture.

5.4.3 Leak Detection

X-ray technology can detect cracks in glass containers like syringes, vials, and cartridges if the spatial resolution of the separation of the glass edges is adequate. If this resolution is sufficient, Xray can detect glass defects under the vial crimp due to its ability to easily penetrate aluminum. This ability also provides options to inspect assembled devices or syringes inside blisters or other secondary packs. For this reason, X-ray detection may be a valuable addition to an automated process or may be used as an offline tool to evaluate the physical characteristics behind a known leaking sample.

Figures 5.4.3-1, 5.4.3-2, 5.4.3-3, **5.4.3-4** show defects identified by X-ray technology.

Spotting the crack in **Figure 5.4.3-2** strongly depends on the angle of the view. For an automated inspection, a minimum of six angles needs to be inspected. A more comprehensive CT scan may prove more valuable for nonautomated, offline sampling, which could be used during an investigation or design study.

X-ray technology also has the ability to detect functional defects in pharmaceutical primary packages, especially in syringes. Many defects can be associated with prefilled syringes, ranging from imperfections in glueing the needle socket to defective luer locks around the syringe socket. Other defects can be associated with bent needles within the needle shield or cut-off needle tips that can cause package

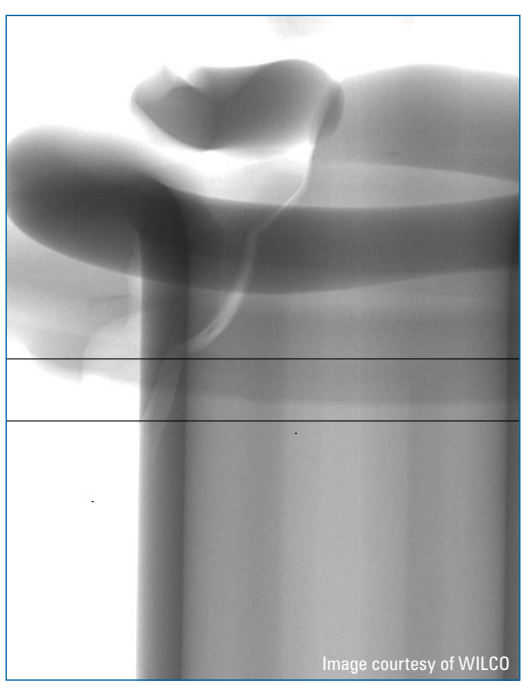

Figure 5.4.3-1 X-Ray Picture of a Large Crack in the Syringe Barrel Inside an Auto-Injector Device

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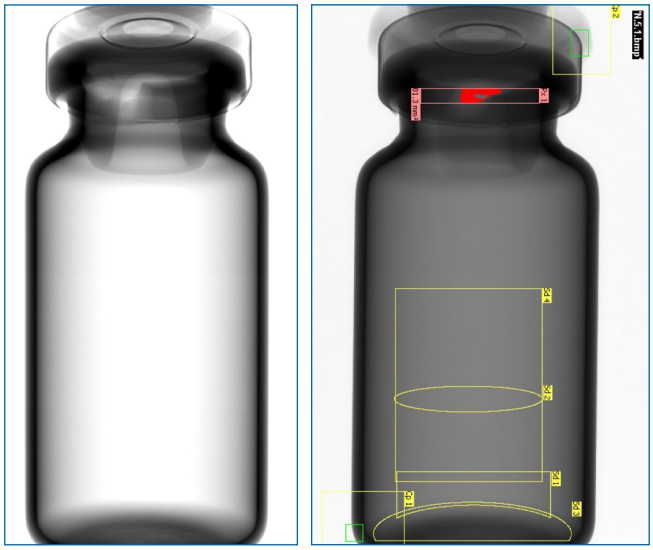

Image courtesy of WILCO

Figure 5.4.3-2 X-Ray Picture of a Vial with a Glass Defect at the Rim under the Crimp (left) and Automated Detection of the Defect (right)

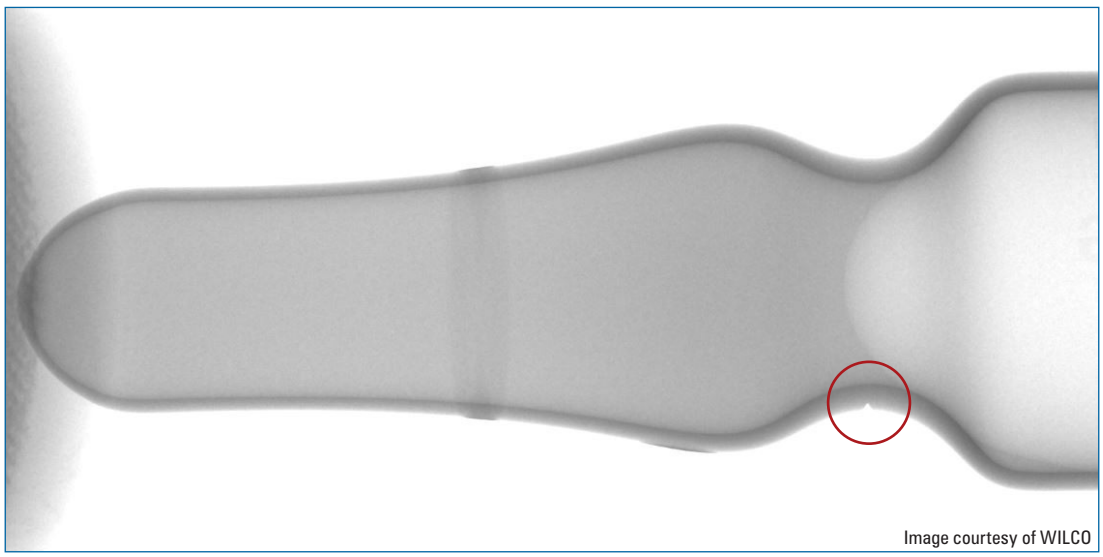

Image courtesy of WILCO

Figure 5.4.3-3 X-Ray Picture of an Ampoule Neck with a Glass Cut Detection of the Defect (right)

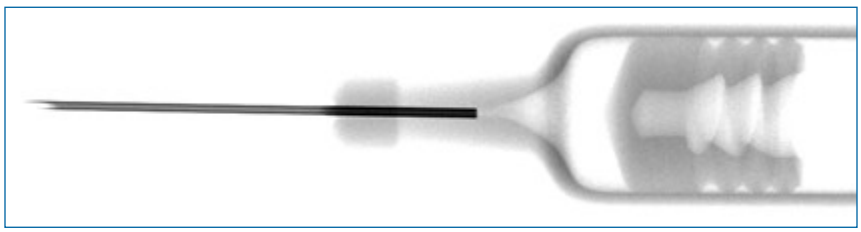

Figure 5.4.3-4 X-ray Picture of Needle Socket with the Glue made Visible

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integrity issues or detrimental effects to the patient, such as pain during administration. Many such defects can be visualized using X-ray.

5.4.4 Considerations and Challenges

An increasingly popular application of X-ray technology in CCI occurs in the package design and development stages. X-ray might be used to support package component selection and assembly parameters, or even to evaluate the impact of storage conditions, such as frozen conditions, on sealing geometries. Low temperature storage is known to contribute to stopper shrinkage as a glass transition state (T_g) is reached; see **Section 4.4**. This effect may be characterized by X-ray imaging. These efforts are typically performed in tandem with a true leak test, such as helium leak detection or headspace analysis.

In a 2016 study, Mathaes et al. performed a comprehensive capping optimization study, evaluating correlations between capping settings, stopper compression, residual seal force, and CCI as tested by helium leak detection. CT scans using X-rays were used to both calculate stopper compression, a critical assembly parameter, and provide visualization of sealing geometries for a range of stoppers ***(20)***. **Figure 5.4.4-1** compares four different elastomers and highlights the ability of CT scans to nondestructively evaluate geometries, including "dimpled" stoppers.

In a 2016 study by Nieto *et al.*, CT scans were used to visually and geometrically explain a difference in leak test results between two stopper groups stored at frozen conditions. CT scans identified gaps between the flip-off cap and the elastomer, as well as the outer edge of the land seal in the group with fewer ideal leak results ***(5)***.

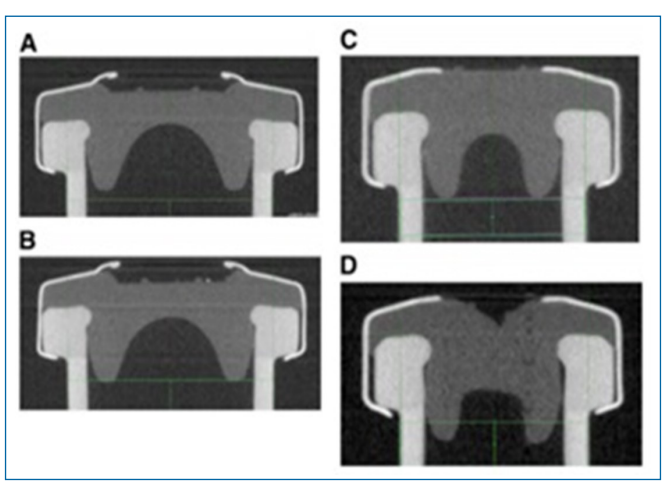

Figure 5.4.4-1 Comparative CT Scans of Four Different Elastomers Used to Calculate Stopper Compression and Fit; Stopper D Shows Signs of Dimpling

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6.0 Additional Considerations for Package Integrity Profiling

A robust package integrity profile for a given product-package system considers factors experienced throughout the product's lifecycle as required by USP <1207>. As such, a package integrity profile must consider anticipated or experienced stresses, including those resulting from transportation and distribution. Such stresses may include shock, vibration, compression, temperature, pressure, and general package storage environment.

6.1 Transportation and Distribution

A package integrity profile is unique to each product-package system, a concept that also applies to storage and distribution considerations. The types of stresses experienced in these lifecycle phases vary widely with dosage form, pack-out configuration, product requirements, and distribution chain. Additionally, the impact of any single stress factor, such as a temperature excursion, can vary with respect to product. An excursion critical to one product may be insubstantial to the quality of another.

Given the potential impact to product quality and sterility, appropriate challenges and assessments should be made to ensure that product quality is not affected during the transportation and distribution circuit. The types of challenge conditions and subsequent assessments performed, however, reflect the properly assessed risks for the system in question.

As the intent of this report is to focus on package integrity, only those distribution challenges impacting integrity are discussed here. Transportation and distribution challenges are not limited solely to integrity of the package, however. A fully integral package containing a proteinaceous product may experience vibration and shock that does not lead to integrity failure but, for example, may cause aggregation of protein. Thus, in addition to post-challenge CCIT, other assessments may be necessary to evaluate the impact of distribution on overall product quality.

6.1.1 Physical Stressors

Perhaps the most obvious challenges experienced during distribution are physical stresses such as shock, vibration, and compression. Often, these types of events result in gross leakage, for instance, a vial shattering. This type of gross defect is easily detected and generally prevents final use of the product. The formation of cracks, checks, chips, and other small defects, however, may result in leakage that negatively impacts product quality but is not readily detectable by the end user. Thus, these types of defects present a more latent and use-specific risk. Furthermore, such types of defects may result in more serious integrity concerns further along in the distribution chain, as a check may develop into a crack, or a crack may widen and permit liquid loss or microbial ingress.

Consequently, to evaluate and reduce risk associated with a shipment, not only the primary package, but the secondary and tertiary packaging should be considered. Numerous standards are available from such organizations as ASTM and the International Safe Transit Association (ISTA) to reliably and reproducibly subject package systems to physical stresses that replicate a variety of distribution chains, including via truck, rail, air, sea, or others.

The effects of temperature on packaging and distribution is also discussed in **Section 4.4** on Cryogenic Conditions.

6.1.2 Pressure

In cases where a low-pressure environment could present challenges to the package and product it contains, evaluation of the pressure differentials in a distribution chain could be critical. This is particularly relevant in cases of air shipment, where product may be stored in cargo areas with partial to nonexistent pressurization that, if left unevaluated, may result in unintended consequences. The ability of a sealed flexible container to resist creep or burst during air shipment is an example of risk that should be characterized.

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Another example, one discussed with increasing frequency as the prevalence of prefilled syringes and prefilled syringe-based devices expands, is the concept of plunger movement. Although the degree to which plunger movement occurs is impacted by fill level, headspace volume, general package design, and pressure differentials experienced, how a low-pressure environment may impact product sterility and quality in these types of systems must also be characterized. As a plunger moves backward, the seal, and possibly the product, may move into an unsterile area of the container. In extreme cases, the plunger may completely exit the barrel.

6.1.3 Testing Approaches

Numerous documents from internationally recognized organizations (e.g., ASTM, ISTA, ICH, ISO) are available for reference and provide reproducible ways of evaluating some of these stressors, including physical shock events, thermal cycling, and pressure cycling. However, the simulated distribution chain should reflect the anticipated stressors. Standards are meant to employ reproducible simulations but may not account for the variations and intricacies of specific distribution networks, such as iterative air shipments or a combination of different modes of transport. As with the application of many test methods and techniques, a proper assessment of the risks should inform decisions on evaluating such risks.

6.2 100% Online Testing

A holistic approach to controlling CCI should be taken to ensure package integrity at the time of manufacture as well as over the shelf life of the product ***(17)***. This holistic approach includes consideration of:

- CCIT method (in-line, near-line, off-line)
- Quality of primary package components
- Primary package design
- Manufacturing process qualification
- Product manufacturing process
- Change control process
- Shelf-life assessment

The CCIT method is a critical part of the control strategy, and an in-line test can make the overall approach more robust. However, the in-line test should not be the sole point of control.

In-line testing is defined as 100% testing of all filled and sealed primary packages for CCI during the production process, after the final sealing operation.

Currently, some regulatory requirements state that all container types fused and filled on the filling line (e.g., ampules, blow-fill-seal containers) must be 100% inspected ***(19)***. For other products, no current regulatory requirement exists for 100% in-line testing. The decision to conduct 100% in-line testing for other products should be based on risk assessment, as part of a holistic approach. A package that is accepted at the in-line test station is not guaranteed to maintain its CCI during the shelf life and use of the product. Conversely, a package that is rejected by the in-line test station is guaranteed not to meet the product's CCI requirements. Therefore, the in-line test station acts as a filter for rejected packages, reducing the risk of CCI failures due to random process failure modes.

Consistent failure modes that are due to equipment malfunctioning or setup can be detected by other means, such as off-line quality control testing. A particular advantage of in-line CCIT compared to off-line quality control testing is the elimination (and cost savings) of handling suspect lots of products due to sample rejection at the off-line test station.

For high-speed lines, current technologies cannot test for the small-sized defects that off-line test methods can. However, since most process failure modes result in medium to large defects in the package, widening the reject limit for in-line testing is a reasonable and low-risk decision. For each product,

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its process failure modes are examined to ensure that the in-line test station LoD does indeed present a low risk for that specific product based on its components, parameters, and requirements.

While it can add value in some situations, technology is not currently available to enable 100% in-line testing for all CCS and product types. Where feasible, regulations and quality risk management principles identified in *ICH Quality Guideline Q9: Quality Risk Management* should be used to determine if 100% in-line testing should be implemented ***(3)***.

Integrating 100% in-line testing into the manufacturing process requires a strong commitment. The in-line test station requires capital investment and the operational cost of the equipment, which adds to the cost of the product. Multiple test stations may be required on each filling line to overcome differences between line speeds and test station throughput. Furthermore, the equipment requires the manufacturer to provide skilled resources to qualify, operate, and support this type of equipment, which is inherently much more complicated than an off-line quality control test station using the same measurement technology.

No one test method fits all products. In-line test methods must be nondestructive for the specific product to be tested.

6.3 Building a Quality by Design Approach into the Container Closure Integrity Testing Program

Pharmaceutical QbD is a systematic approach to development that begins with predefined objectives and emphasizes product and process understanding and control based on sound science and risk management. A QbD concept can be built into a CCIT program, including such elements as:

- Quality target product profile that identifies the critical quality attributes (CQAs) of the product-package system
- Package design and understanding, including identification of critical material attributes that can affect package integrity
- Process design and understanding, including identification of CPPs, linking the material attributes and process parameters to the CQAs
- Control strategy that includes specifications for the product-package configuration with respect to CCI as well as controls during each step of the manufacturing process that will help meet those specifications
- Regular quality checks on process capability and continual improvement initiatives

One of the ways users can build QbD into a CCIT program is to employ a risk-based approach. A risk-based CCIT program is built on science-based decisions; it offers an ongoing database of product lifecycle CCIT results (also called a package integrity profile) and a risk management tool for package integrity assurance throughout the lifecycle of the product. The program should demonstrate CCI as a function of ongoing, operative variations that takes into consideration:

- Design and material of the package
- Package assembly
- Processing conditions
- Storage, distribution, and stability conditions

Following are some key aspects that can be used to build a risk-based CCIT program by considering QbD principles:

- **CCI over the entire lifecycle of product**: The CCIT tool kit needs to demonstrate CCI over the entire lifecycle of the product. More than one test may be employed during a given product's lifecycle based on the critical parameters to be controlled at that stage in the lifecycle of the product-package system. **Section 6.4** discusses the lifecycle approach related to bulk flexible containers.

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- **Choice of deterministic versus probabilistic methods**: Deterministic methods are those where the leakage event is based on phenomena that follow a predictable chain of events. Such methods require less preparation, are performed with instruments, and offer quantitative outcomes. Probabilistic methods offer qualitative outcomes that are subject to the probability of a series of events to occur but may still provide valuable information when properly applied. For example, some probabilistic methods may provide information on the location of a defect in a flexible pharmaceutical container. Some probabilistic methods may prove more advantageous than deterministic methods with respect to enhanced sensitivity. For example, internal-pressure testing has in some cases demonstrated sensitivity down to a smaller micron size when compared to the sensitivity established with a deterministic method.

- **Choice of off-line versus in-line, on-line or at-line test methods:** Implementation of a chosen CCIT method may vary in its nature depending on the stage in the product lifecycle and the CQAs pertaining to package integrity required at that specific time point.
  - **Off-line methods:** Measurement does not involve samples removed directly from the manufacturing line. Test methods are typically not high speed, are not integrated into the manufacturing line, and are usually implemented in a laboratory setting.
  - **In-line methods:** Measurement where the sample is not removed from the process stream and methods are not destructive to the sample (see **Section 6.2**).
  - **On-line methods:** Measurement where the sample is diverted from the manufacturing process and may be returned to the process stream; these test methods are not destructive to the sample (see **Section 6.2**).
  - **At-line methods:** Measurement where the sample is removed, isolated from the process stream, and analyzed.

  In-line and on-line test methods are typically needed when 100% testing is required during manufacturing. At-line and off-line tests are better suited for testing based on a scientifically valid sampling plan, stability study, and development environments. The selected method(s) must be suitable for the intended use and scope of the specific CCIT. In-line and on-line methods can potentially provide greater assurance that all packages have integrity and can yield instant feedback in the event of package misassembly.

- **Sampling:** Scientifically valid sampling plans should be based on risk assessment in association with appropriate statistical criteria. This could lead to various scenarios, including one where the occurrence of leaks is sufficient to require 100% testing due to a complex or highly variable manufacturing process. In cases where the process is under statistical control, sampling plans may be based on standards adhered to within the local quality management system.

  Factors to consider for sampling plans include the complexity of the product-package design and related manufacturing process, available test methods required to ensure product quality, and prior experience with similar configurations.

- Regardless of the nature of the test method (deterministic or probabilistic), the number of samples can provide sufficient confidence through risk assessment.

- **Defining the MALL:** CCI is established when a package meets its maximum allowable leak limit. Since product quality requirements define the MALL, the choice of test method may be based on the stage of the product lifecycle (initial development, routine manufacturing, shelf life, and stability assessments) and should depend on the critical parameters to be controlled at the specific stage.

- **Designing a CCIT tool kit:** The CCIT tool kit should encompass different test methods that can be employed to detect defect types critical to product quality and sterility. Ideally, the tool kit

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would provide the user with not only details about the test method, its application, and leak detection capability, but also with testing efficiency, including testing speed (time taken to detect leaks of critical value) and throughput (rest times for sample equilibration and measurement of control groups). This information will help the user determine which CCIT can be implemented at each stage of the product lifecycle based on the desired CQAs to be achieved.

- **Ensuring continuous improvement:** In the event of a change to the package design, material, or manufacturing processing conditions (including sterilization conditions), the risk-based strategy should be designed to trigger a review of the CCIT tool kit to ensure it includes test methods that can detect defects critical to product sterility and quality.

6.3.1 Risk Assessment

A risk-based approach using the quality tools mentioned in ICH Q9 can be a very effective way to assess sterile package integrity and testing applications. Following the lifecycle phases, relevant package integrity parameters are considered to determine product-package quality intent and how to evaluate the critical parameters that effectively impact the integrity of the CCS.

A lifecycle assessment process provides ample time and opportunity to develop improved learning, better understanding, and ongoing generation of compelling data to support the risk of package integrity failure. This QbD approach consists of learning, evaluation, and control strategy determination. The impact or risk of an integrity failure is relevant to both patient and product. The complexity of a package integrity risk assessment is derived from the numerous inputs to the manufacturing and packaging processes and the multiple ways of evaluating the outcomes of these processes.

A first step in risk assessment (e.g., using the Hazard Analysis Critical Control Points (HACCP) quality tool) is to identify each incremental process step and the appropriate materials relevant to that step. Next, a verification of the potential risk of failure (i.e., a leak) at the process step is made and a list drawn up of causal factors that might lead to each failure type. Another list is developed that indicates what controls may exist that could mitigate the risks. Subsequently, the risk tool can be used to determine what types of testing (monitoring) are valuable, how frequently testing should be performed, and where or what types of samples should be taken for evaluation.

Similar to Six Sigma thinking (define-measure-analyze-improve-control), the risk assessment process is an ongoing methodology for quality improvement.

6.4 Bulk Container Lifecycle Approach

Bulk containers go through two distinct phases of lifecycle: First, the development phase that encompasses material selection, design, and qualification and, second, the product phase, from manufacturing steps through supply chain to point of use. QbD and QRM principles can be applied throughout both phases, during which specific risks are identified and evaluated using appropriate methods. The responsibilities and capabilities of identifying and mitigating such risks can be divided between the development and manufacturing centers by the supplier as well as the user before, during, or after use.

Increasingly, according to the criticality of their application, users are asking suppliers to conduct testing, or choose to do it themselves to mitigate the risk of leakage or contamination. The materials used for these applications are engineered polymers that must be specified and controlled appropriately up the supply chain to ensure consistency of product quality. Therefore, prior to implementing specific testing, a consistent product robustness must be established using QbD principles and a QRM approach that will result in an appropriate validation and process control strategy. **Figure 6.4-1** illustrates an initial framework to manage the current options.

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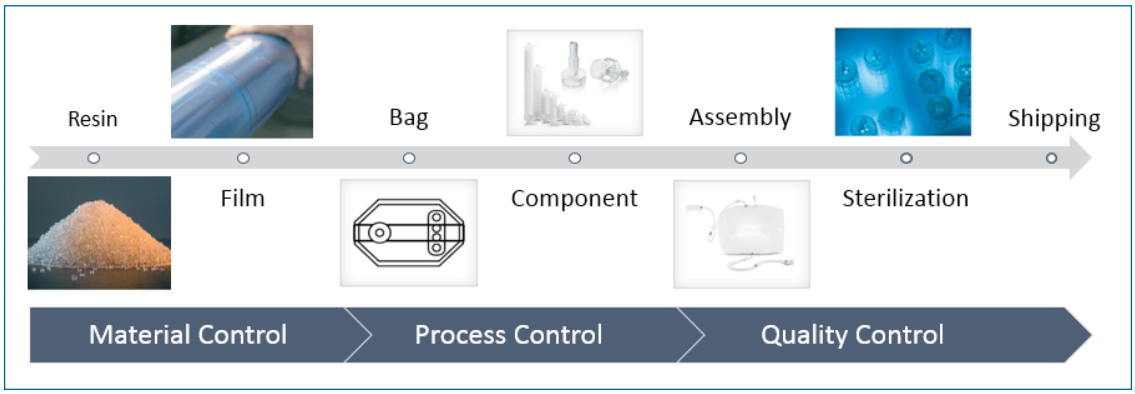

Figure 6.4-1 Single-Use System's Lifecycle at Manufacturing Site by User are Items Similar to those in Table 6.4-1

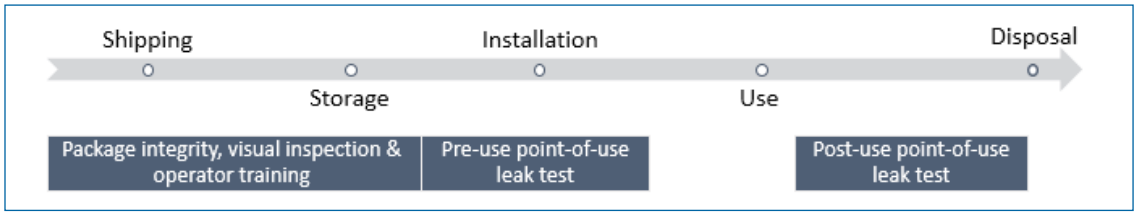

Figure 6.4-2 Single-Use System's Lifecycle at User's Site

Table 6.4-1 Considerations for Single-Use System's Lifecycle at User's Site

Area to Manage Product Quality	Considerations
Risk assessment	Evaluate the quality risk from the supply chain as well as the risks associated with their processes that can affect integrity; develop a control strategy, which may involve use of a physical leak test method
Incoming goods inspection	Identify seal quality and integrity attributes to be evaluated upon receipt
Assembly storage	Ensure that products are stored in a manner that prevents damage
Assembly unpacking and installation	Ensure that assemblies are transported through the plant in a manner that does not induce damage; support the product appropriately (large surface), eliminating use of sharp tools and short-term storage racks with sharp edges
Visual inspection	Check visually for marks, damages, and abnormalities on the SUS
Pre-use leak testing	Verify integrity after transportation and handling directly before use; supplement release testing at supplier
Use	Use the SUS within its specifications
Post-use leak testing	Verify integrity of the SUS after use to support the sterility claim on the final product
Operator training	Ensure that operators understand in which modes assemblies may be damaged by handling and deployment, how to inspect an assembly prior to use, and how to run a leak test
Supplier feedback	Inform supplier of integrity failures so supplier can understand and correct issues

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For the flexible container manufacturer, test methods may differ between the phases of the lifecycle. At the supplier site, specific test methods are used during the design phase to establish the capability for integrity assurance. The right materials in a laminate structure are selected and the welds and seals are designed for the intended use. Other tests are applied during the manufacturing phase, such as those used for in-process controls.

6.5 Educational Simulation about Limits of Detection and Method Selection

In this textual illustration of the key role leak testing can play in the lifecycle of a parenteral product, fictional firm PharmaCo-A is working to bring a new parenteral product to market. During development, the inherent integrity of the primary package met the MALL for the intended product. However, during scale-up, poor quality package components and line equipment issues resulted in defective packages. Part of the action plan to ensure the absence of such rejects includes incorporating leak testing equipment into routine manufacturing to monitor for and, if necessary, cull out defective product-packages. This methodology is also intended to support quality assurance studies. Details reveal how four leak-test technologies were screened, optimized, and validated and how the LoD was determined.

This illustration is purely fictional, and the content is provided for educational purposes only. There is no requirement to follow this approach in content, specificity, or scale. Any similarity to an actual product, package, or leak test method is unintended and coincidental.

- **Package development.** PharmaCo-A was trying to bring a new parenteral product to market. The CCS chosen was based on a variety of package selection criteria. The inherent package integrity required that the system meet the MALL set for the intended product. To check this, empty package units were assembled and processed on mock-production equipment and the leak tested using a highly sensitive quantitative test method. Findings showed that, when properly assembled, the gaseous leakage rates of the packages met the MALL, ensuring proper product containment without risk of product contamination. These studies provided valuable information on optimum component design, component processing, and package assembly parameters that PharmaCo-A used during the next phase of package processing and assembly validation.

- **Package processing and assembly validation.** Later in the course of manufacturing scale-up, the potential for cracked containers, torn closures, and misassembled container closures was noted. Failure mode analysis implied such defects were linked to insufficient process control, outdated production equipment, and a lapse in component quality. PharmaCo-A put in place an action plan that addressed each of these issues to minimize the risk of creating defects during routine manufacturing. In addition, the company decided to identify and implement a nondestructive leak test method for monitoring batch quality and for culling out problematic packaged product units when necessary. PharmaCo-A planned to analyze the resultant data for trends as part of their ongoing, package-integrity profile database.

- **Leak test method selection.** One requirement for the selected leak test method is the need for compatibility with testing product-filled package units in routine manufacturing. PharmaCo-A also chose to employ a nondestructive test method so the actual tested product could be marketed or used for other testing purposes. The test must accurately screen out defective packages, such as those observed during scale-up. Other test method requirements related to test speed, equipment footprint, and more may be evaluated. PharmaCo-A intended to employ this same method for verifying the package integrity of samples retained for quality assurance stability studies, if possible.

- **Leak test method qualification.** PharmaCo-A identified four leak-test technology candidates. Four instruments were leased and installed at their process research and development site. Each instrument was qualified for use using appropriate calibration equipment and tools.

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- **Leak test method feasibility.** Feasibility studies were performed to investigate each method's potential for detecting the various defects observed during scale-up. This work was done using a population set of nondefective units (negative controls) plus defective units (positive controls). Positive controls included rejects culled from manufacturing scale-up studies, supplemented with artificially simulated rejects (type defects). A small number of packages made defective by laser-drilling through the package wall also were included. The outcome of these screening studies proved all four technologies had the potential to identify packages having larger defects.

- **Leak test method optimization and validation.** PharmaCo-A proceeded to optimize test method parameters for each of the four technology candidates. Test package units for this phase included negative control packages, a small set of type defects, along with a much larger pool of laser-drilled defective packages. The larger defect subset included defects of various sizes placed in various package locations.

  The goal of optimization was to achieve maximum defect detection sensitivity, while minimizing the risk of falsely rejecting good units. Method optimization studies also investigated each method's performance for possible variability in leak test equipment setup and operating parameters, package dimensions and materials, and testing environment (e.g., humidity). Thus, each leak test method was evaluated for specificity (ability to detect leaking packages despite potential interfering factors) and robustness (ability to detect defects despite shifts in test parameters, test equipment operation, test environment, and test units). The detection range (largest leak sizes and types detectable) was also evaluated. All four methods performed satisfactorily and comparably.

- **Leak test method detection limit and accuracy.** PharmaCo-A performed a study to establish and compare the detection limit and accuracy of the four potential leak test methods. PharmaCo-A used a population of negative control product-filled packages that had no known leaks of concern and a population of positive control product-packages. Unlike earlier studies that employed more qualitatively defined type defects, all positive control defects for these studies were laser-drilled and sized by gas flow-rate methodology. Defect sizes ranged from the smallest leaks that could be reliably created (2 μ in nominal diameter) to larger-sized leaks (20 μ in diameter). PharmaCo-A chose laser-drilled defects because they most closely represented the smallest defects likely to occur, they could be consistently manufactured, and their size could be independently certified.

Each test method was evaluated over the course of three days by two or more operators. Only one instrument was used for this initial study (additional work was planned to check for intermediate precision using multiple instruments once the final technology was chosen). New positive control subset populations were used for each test day and each test method to lessen the risk of defect-size change caused by repeated testing and handling. All methods were considered nondestructive to the package; therefore, the same negative control population set was tested throughout. Positive and negative controls were tested in random order in a blinded fashion. Each operator was instructed to perform a leak test using a known good package (a "master") following any failing (leaking) result. Testing was allowed to continue if the master test result passed (no leakage detected), thus confirming satisfactory instrument performance.

PharmaCo-A predefined the test method detection limit as that smallest defect size (defined as the average size in each positive control-size subset) in which 95% or more of the defect size-subset was detected as leaking on all test days. All defects larger than the detection limit must fail (leakage detected). Method accuracy was confirmed if all negative controls pass (no leaks detected).

Detection limit and accuracy validation study data for the four test methods are provided in **Table 6.5-1**.

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Table 6.5-1 Leak Test Method Detection Limit and Accuracy Determination Example*

Nominal Leak Path Diameter (n = units tested per test, per day)	Test Packages Rejected as Leaking (%)											
	Test Method A			Test Method B			Test Method C			Test Method D		
	Day 1	Day 2	Day 3	Day 1	Day 2	Day 3	Day 1	Day 2	Day 3	Day 1	Day 2	Day 3
0 µm (n = 300)	0.3	0	0.7	0	0	0	0	0	0	0	0	0
3 µm (n = 40)	90	87.5	92.5	5	7.7	5	30	0	20	0	0	2.5
6 µm (n = 40)	100	100	95	12.5	20	10	100	2.5	0	2.5	2.5	2.5
10 µm (n = 40)	100	100	100	20	10	30	100	100	95	50	20	20
20 µm (n = 40)	100	100	100	100	100	100	100	92.5[1]	100	80	90	100
Leak Detection Limit (µm)	6 µm			Data suggest 20 µm			10 µm			Undefined		
All Larger Defects Detected (Yes/No)	Yes			Data absent			Confirmed			Undefined		
All Negative Controls Pass (Yes/No)	No			Yes			Yes			Yes		

* Data are hypothetical

[1] Later investigation found all 3 packages rejected to be clogged. 100% of the remaining defects were rejected.

The tabulated data results are discussed below.

- **Test Method A.** All positive controls in the 6 µm subset were detected at least 95% of the time on all three test days. All larger defect units were detected on all test days. Therefore, a test method detection limit of 6 µm was confirmed.

  Method A failed to accurately identify all negative controls on all test days; however, a small chance of falsely rejecting good packages was observed. Thus, the predefined test method acceptance criterion for method accuracy was not met.

- **Test Method B.** The smallest defect subset in which 95% or more of the positive control units were detected on all three test days was 20 µm. Study acceptance criteria also required confirmation of the method's ability to screen all defects larger than the apparent detection limit; however, no larger defect size dataset was included.

  All negative controls passed on all test days, thus confirming method accuracy in differentiating between no-defect packages and packages having defects 20 µm in nominal diameter. The method's accuracy in screening defects larger than the detection limit was unconfirmed.

- **Test Method C.** The smallest defect size subset in which 95% or more of the positive control units were detected on all three test days was 10 µm. All negative controls passed.

  PharmaCo-A's detection limit acceptance criteria also required the detection of all positive controls with defects larger than the detection limit. In this case, the method detected only 92.5% of defects larger than 10 µm on Day 2 (three packages failed to be detected); therefore, the detection limit could not be confirmed.

- **Test Method D.** All negative controls passed. However, the method was unable to reliably detect 95% or more of the defects in any positive control subset on all three test days. Therefore, no detection limit could be established, and method accuracy remained unproven.

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PharmaCo-A further evaluated and compared study results, as described below.

- **Test Method A.** This method was notably best at culling out defective packages of all defect sizes; however, the observed small risk of falsely rejecting good packages was cause for concern. Two paths forward were proposed:
  - First, adjust test method parameters to reduce method sensitivity to a small degree, thereby reducing the false reject risk. While this would likely compromise the method's ability to detect 3 µm defects, data suggest it would have little impact on the current 6 µm detection limit.
  - Second, use the method as-is to ensure capturing as many defects as possible, knowing that some good product may be lost. Quality assurance stability samples tested in this way would also run a greater risk of false positive results, potentially triggering costly investigations and marketed-product recalls. To address this concern, quality assurance protocols were proposed to require replicate testing on each sample to confirm findings, including instrument performance verification steps should any test package fail.

PharmaCo-A continued to pursue Method A and explore the options proposed.

- **Test Method B.** Unlike Method A, Method B showed no tendency to falsely reject good product. The detection limit defect size was much larger than Method A, however, so smaller defects capable of putting product quality at risk would likely be missed. Given the availability of a more sensitive testing option, further work on Method B was placed on hold.

- **Test Method C.** The 20 µm reject findings of Day 2 were considered suspect. A follow-up investigation revealed the three defects missed were clogged; therefore, these data were discounted. This information allowed a detection limit of 10 µm to be defined, and the method's accuracy to be confirmed. However, the wide discrepancies in leak detection observed for the smaller defect subsets among test days caused PharmaCo-A to question Method C's overall reliability. Further work on this method was placed on hold.

- **Test Method D.** As no detection limit could be defined, Method D was considered too insensitive for PharmaCo-A's use and was therefore eliminated from further consideration.

Leak test Method A was finalized and put into place to support routine manufacturing and to verify package integrity of quality assurance stability study samples. PharmaCo-A obtained commercial product market approval. Ongoing Method A leak-test data, along with package component quality trends, remained part of the package integrity profile database. The information gleaned helped ensure ongoing marketed-product package integrity.

7.0 Conclusion

Best practices in pharmaceutical packaging suggest that all activities must be evaluated for risk to the product quality attributes and, by extension, to the patient. Failure of CCI is a critical risk that can potentially result in serious to dire consequences. While USP <1207> defined the basic concepts of CCI and integrity testing methodologies, this technical report serves as a resource that addresses the challenges with test methods and complex packaging systems, introduces new uses of some existing technologies, and discusses additional considerations, such as use of a risk-based approach relevant to the packaging lifecycle. There is no one-size-fits-all approach for packaging integrity testing methods; however, using a risk-based approach and leveraging existing test methods as described in this technical report can assist in better understanding and potentially mitigating, if not eliminating, the risks to package integrity.

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